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Evidence from a Major Trading Glitch



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Slow-Moving Capital and Execution Costs: Evidence from a Major Trading Glitch

Abstract

We investigate the impact of an exogenous trading glitch at a high-frequency market-making firm on standard measures of stock liquidity (spreads, price impact, turnover, and depth) and institutional trading costs (implementation shortfall and VWAP slippage). Stocks in which the firm accumulates large long (short) positions increase (decrease) by about 4% during the glitch and become substantially more illiquid. It takes one day for prices and spread-based liquidity measures to revert. Institutional trading costs, however, remain significantly higher for more than one week. Both liquidity measures are also weakly correlated outside the glitch period, suggesting they capture different aspects of liquidity.

Keywords: Slow-Moving Capital, Market Making, Liquidity, Algorithmic Trading, Institutional Trading Costs.

JEL Classification: G10.

1. Introduction

When buyers and sellers arrive asynchronously, there is a role for providing immediacy (Demsetz, 1968; Grossman and Miller, 1988). If capital is slow-moving then shocks to the inventory and risk-bearing capacity of immediacy providers should affect market liquidity (Amihud and Mendelson, 1980; Duffie, 2010). This is challenging to test empirically, however.

First, it is difficult to observe exogenous shocks to the inventory of a market participant. For instance, Hendershott and Menkveld (2014) rely on a statistical state space model to separate temporary price pressure due to inventory shocks from the permanent price impact due to information. It is also difficult to measure exogenous shocks to the risk-bearing capacity of individual liquidity providers. Extant empirical literature finds evidence that shocks to the inventory levels of NYSE specialists are related to trading costs, but such shocks could also be related to new information or changes in firm risk (Hendershott and Seasholes, 2007).¹

Second, it is not clear how one should measure market liquidity. Most existing papers use some measure of the quoted spread or of the effective spread, often decomposed into realized spread and price impact, to measure market illiquidity and its components (trading and order-processing costs, inventory costs, and adverse selection). But as pointed out by Domowitz et al. (2005) and others, the quoted spread only represents the inside quote for a small trade size. The effective spread improves on that as it is not restricted to the inside quote. However, it is still only an ex-post measure for a specific (typically small) trade size and does not correspond to an ex-ante measure of liquidity of a potential (large) trade size. Thus, especially for large institutional investors, it is not clear that it is an appropriate measure of liquidity. Indeed, large institutional investors typically rely on measures of implementation shortfall at the parent-order level, since their trades are broken down into multiple small trades (Perold, 1988).

In this paper, we shed light on the importance of inventory and capital shocks by examining the impact of a major trading glitch at a large high-frequency market-making firm (Knight Capital, KC henceforth) on different measures of liquidity. The glitch—originating from the erroneous implementation of a trading software—occurred on August 1, 2012 during the first thirty minutes of trading and resulted in numerous erroneous trades on a set of NYSE-listed stocks.

Using public data, we identify the set of stocks affected by the trading glitch and measure the cost of trading the affected stocks on the day of, as well as subsequent to, the

¹To try to control for such endogeneity, Comerton-Forde et al. (2010) look at mergers between broker-dealer firms as a positive shock to their market making capital.

glitch. We use two measures of trading costs. First, we use a proprietary execution data set from a large investment bank that contains measures of implementation shortfall (IS) and the slippage against the realized volume-weighted average price (VWAP Slippage) for a large number of institutional investors parent-orders that are broken down into several children-orders and traded using standard execution algorithms. Second, we compute standard market microstructure measures of effective spread (decomposed to realized spread and price impact), turnover, and depth.

Since the trading glitch was due to a technical problem that became quickly common-knowledge (Reuters published its first news feed around 10:00 AM on the day of the glitch), the dramatic long and short positions accumulated by KC on the glitch-affected stocks were unrelated to fundamental information on the underlying stocks themselves. Thus, they constitute a natural experiment to test the effect of exogenous inventory and capital shocks on market prices and liquidity.

First, we show in Figure 1 that stocks that were bought as part of the glitch experience an average positive jump of 4% concurrent with the heavy glitch-induced trading volume. About half of the jump reverts by 10:30 AM when news of the glitch has become public, but it takes until market close before prices revert to their pre-glitch levels. The price-path followed by stocks that were sold as part of the glitch is the mirror image of that of bought stocks: they initially drop by about 4%, recover half of the drop by 10:30 AM, and revert fully by 4 PM. Interestingly, we find that both groups of stock over-shoot their ‘equilibrium’ level around 3 PM which correlates with a significant increase in trading. Overall, the picture resembles the price patterns generated by the slow-moving capital model of Duffie (2010) in response to an unexpected (positive or negative) supply shock, which takes a full day to be absorbed by the market, even though it is unrelated to any fundamental news.² Qualitatively, the evidence is consistent with the price drop and reversal documented by Coval and Stafford (2007) and Edmans et al. (2012) around mutual fund fire sales. Quantitatively, however, for a similar magnitude return drop the fire-sales literature finds a much longer reversal time of close to two years.³ Taken at face value, these papers suggest very slow-moving capital. In his discussion, Duffie (2010) suggests that this slow reversal might be the consequence of the difficulty in markets to separate a pure supply shock from an informationally-motivated

²It is plausible that prior to 10:00 AM, there was some uncertainty about the motive for the heavy buying and selling. However, after 10:00 AM the glitch becomes public news. Further, in contrast to what occurred around the ‘flash-crash’ event, for example, there was no contemporaneous abnormal move in the VIX during this episode, which suggests that markets were not ‘spooked’ by the glitch event.

³Coval and Stafford (2007) find a 15% cumulative return drop due fire sales that starts 12 months before it begins to revert and takes more than 18 month to revert (see their Figure 2). Edmans et al. (2012) find a 5% return drop around large unexpected fire-sale orders that takes more than 24 months to revert (see their Figure 2).

trade. Duffie conjectures it might take the market several months to learn that the fire-sales, identified as such ex-post by researchers, were not, in fact, informed trades. Our setup circumvents this issue and, consistent with the above view, our results suggest that if the market quickly learns that the trade is not informationally motivated, then the reversal is a lot quicker.⁴

Having established that the impact of the unexpected (negative and positive) supply shocks on the level of prices is significant and takes one full day to revert, we investigate how stock liquidity is affected by the glitch. We expect two types of effects. First, the trading glitch allows us to test the impact of exogenous supply shocks in individual stocks on their liquidity. KC had to rapidly adjust its inventory to satisfy margin calls. An increase in KC's trading intensity to unload positions should increase trading costs (e.g., Barra, 1997). Importantly, we conjecture that capital market imperfections such as slow-moving capital and predatory trading are likely to accentuate the increase in trading costs. In particular, we control for well-known determinants of liquidity such as volume and volatility in our empirical tests.

Second, the inventory shocks together constitute a major shock to the risk-bearing capacity of KC. Losses on KC's position were estimated at around \$460 million and led to KC eventually being bought up by a competitor (see Section 2 for details). Since KC was a so-called designated market maker (DMM) for a set of NYSE stocks, we would expect, if market making capital is scarce and slow-moving, that such a shock would affect the market liquidity of all stocks for which KC was a DMM, whether or not they were heavily traded as a result of the glitch itself.

To distinguish between the 'supply shock' and the 'market making capital' channels, we report the impact of the glitch on our two sets of trading cost measures for stocks that experience an abnormally high trading volume during the glitch (glitch-affected stocks) and for stocks that experience a normal trading volume during the glitch but for which KC was a designated market maker (DMM stocks).

Our main findings are as follows. First, all transaction cost measures on glitch-stocks experience dramatic increases on the day of the glitch. Institutional trading cost measures increase substantially on the day of the glitch. Strikingly, this increase remains statistically and economically significant for more than one week after the glitch. Importantly, institutional trading costs increase despite controlling for many of their known determinants, including the participation rate. This result is consistent with market frictions that affect trading costs beyond standard determinants following large supply shocks. Because of the

⁴A recent paper by Wardlaw (2019) also finds that the pure price pressure effect may be a lot shorter and less important than measured in these previous papers.

persistent increase in trading costs, the impact of the trading glitch on the cumulative increase in institutional trading costs is substantial. In a back-of-the-envelope calculation we estimate that the total cost of the glitch to institutional traders on glitch affected stocks is on the order of \$100 million. In contrast, standard microstructure measures (spreads) revert quickly towards pre-glitch levels, suggesting that market liquidity is quite resilient.

The evidence on institutional trading costs is consistent with inventory shocks having a large and persistent impact on liquidity. To test for the stock-specific ‘supply shock’ channel, we distinguish between institutional trades that seek an execution in the same direction as the expected supply shock based on the glitch. As an example, consider a stock that KC bought as a result of the glitch. Then we expect that KC (or rather the intermediary who bought the position from KC) will seek to offload the position in the days following the glitch. Thus, the persistent supply shock hypothesis would suggest that an institutional trade to buy would get more favorable execution than a trade to sell, since the latter would be on the same side as that desired by KC (or the intermediary holding the KC position). We find strong evidence that these ‘same side’ trades have persistently higher execution costs than ‘opposite side’ trades.

As evidence for the ‘market making capital’ channel, we find that stocks for which the firm was a DMM experience a significant increase in their institutional trading costs on the day of the glitch, even though these stocks were not heavily traded during the glitch. Furthermore, we also find that effective spreads on these stocks that are not in the S&P 500 increase significantly in the aftermath of the glitch. Since KC suffers a capital shock, its incentive to provide liquidity drops as predicted by the market making capital theory. The decline in liquidity is more persistent for stocks that are less actively traded.

[Clark-Joseph et al. \(2017\)](#) use an unexpected trading halt on the NYSE as a natural experiment to show that DMM participation lowers spreads. Our DMM analysis differs in that we focus on a single DMM and directly link a shock to its capital to the liquidity of the stocks it supervises. Further, since the trading halt affected the whole NYSE trading system (along with the DMMs), the natural experiment in [Clark-Joseph et al. \(2017\)](#) cannot causally link DMM participation and stock liquidity. [Bessembinder et al. \(2019\)](#) exploit a discontinuity in DMM’s obligations to provide liquidity in a specific security based on whether its past-month consolidated volume is above or below 1 million shares (NYSE Rule 104). Their results support the positive influence of DMM participation on liquidity. Our study mainly focuses on S&P 500 stocks (i.e., the set of stocks for which we have execution data), none of which is on the lower side of the 1 million shares threshold during our sample period.

Using NYSE data from 1994 to 2004, [Comerton-Forde et al. \(2010\)](#) show that a specialist’s

inventories and trading revenues are positively related to the value-weighted spread among stocks *assigned* to that specialist. Our paper supports their findings but differs on several key dimensions. Most importantly, the supply shock channel argues that a shock to the inventory of a market participant in a given stock affects the liquidity of that stock, independently of whether the market participant is a market maker in that stock. Since most glitch-affected stocks do not have KC as DMM, we can make this distinction. In addition, the glitch represents a natural experiment to evaluate the effect of a large capital shock to a market maker. As pointed out by [Comerton-Forde et al. \(2010\)](#), common factors make it difficult, in general, to disentangle between the role of inventories and revenues for future liquidity.

Finally, we perform additional tests to understand why traditional spread-based measures in the microstructure literature differ strikingly from institutional trading cost measures, especially with regards to measuring the persistence in illiquidity following the glitch. We show that spread-based measures exhibit only little relation to the institutional trading cost measures in our data. Although this discrepancy has been noted in the prior literature (see e.g., [Jones and Lipson, 2001](#)), the novelty in our analysis is that we can employ a side-by-side comparison of spread-based and institutional trading cost measures by comparing them over the same trading interval and controlling for many determinants of trading costs. Such comparison is not possible in typical institutional trading datasets (e.g., Abel Noser data) as exact start- and end-time of the parent-order execution are not available to the researcher and consequently it is not possible to control for the market conditions during the execution horizon.⁵

We then consider additional measures of liquidity and trading activity to understand what drives the discrepancy between cost measures. First, we find that turnover of glitch-affected stocks increases significantly on the day of the glitch and remains abnormally high during the week following the glitch. This increase in trading activity is consistent with the supply shock explanation and is not picked up by spreads. Second, we analyze market depth on glitch-affected stocks and find that it decreases significantly in the week following the glitch. These results highlight the multi-faceted nature of liquidity, some facets of which may be more relevant for institutional investors. This calls for additional research to better understand the discrepancy between the different types of liquidity measures.

Our findings are broadly consistent with theories of ‘slow-moving capital’ ([Duffie, 2010](#)). Interestingly, liquidity is adversely affected with more persistence than prices. This suggests fruitful avenues for theories that disentangle between price shocks and liquidity shocks. Our findings also show that liquidity provision by high-frequency traders, inasmuch as it generates

⁵In line with our results, [Eaton et al. \(2019\)](#) document that, in contrast to spreads, institutional trading costs are not markedly affected by decimalization.

more instances of ‘flash-crashes’ similar to the ones studied in this paper, may have adverse consequences for market participants, such as institutional investors, that are difficult to detect using traditional measures of market liquidity.⁶ In this respect, our analysis is relevant for liquidity research and is not restricted to glitch-specific events. The glitch simply provides a natural experiment to test theories of supply shocks and market making capital.⁷

The rest of the paper is organized as follows. Section 2 provides some background information on the trading glitch and discusses the relevant theories for our main empirical analysis. Section 3 describes the liquidity measures used in the paper. Section 4 presents the data and methodology. Section 5 discusses the impact of the glitch on liquidity. Section 6 investigates why standard liquidity measures differ from institutional liquidity measures. Section 7 details several robustness tests. Section 8 concludes.

2. The August 1, 2012, Trading Glitch

This section describes the Wednesday August 1, 2012, trading glitch and discusses its potential impact on stock liquidity.

2.1. Timeline of Events

On August 1, 2012, Knight Capital (KC) experienced significant issues with its automated order routing system. Over the previous days, KC had updated its servers so that its customers could benefit from the introduction of a new retail liquidity program on the NYSE. However, one of KC’s servers was not updated correctly. The server did not recognize when an order had been executed and therefore kept sending new orders to the market. Because of the system’s complexity and a lack of adequate control procedures, KC’s technology team was not able to identify the source of the problem immediately. As a result, the glitch made KC’s system send millions of orders to the market over the first 30 minutes of trading. Consequences were substantial. At some point, KC held a \$3.5 billion net long position in 80

⁶There are numerous empirical studies documenting that HFT activity is positively correlated with improvement in market quality measures; see, for instance, [Hendershott et al. \(2011\)](#) and [Brogaard et al. \(2014\)](#). [Kirilenko et al. \(2017\)](#) examine the trading patterns of the HFTs during a large and temporary selling pressure empirically and find no marked differences compared to normal periods.

⁷Even though this paper is not concerned with a study of glitches in general, it is worth emphasizing that glitches are common in today’s computer driven markets. Starting with the Flash Crash on May 6th, 2010, financial markets have experienced many similar crashes in different asset classes and countries. For example, major crashes occurred in the German DAX index on August 18, 2011, Indian NSE index on October 5, 2012, U.S. Treasury market on October 15, 2014, and the cryptocurrency Ethereum on June 22, 2017. Further, stock-specific mini flash crashes are prevalent in the U.S. equity markets: [Johnson et al. \(2013\)](#) report that there have been 18,520 mini flash crashes between 2006 and 2011 with durations less than 1.5 seconds.

stocks and a \$3.15 billion net short position in 74 stocks. These positions led KC to realize a \$460 million loss.⁸

To get a sense of when market participants might have learned of the trading glitch at KC and identified it as a “non-fundamental shock,” we review in the Internet Appendix all the public news articles that appeared on Bloomberg and mentioned the trading glitch. The first relevant Bloomberg article appeared at 9:48 AM. It mentions abnormal trading in the KC stock (whose price dropped 5.1% during the first trading hour), which suggests that by that time some market participants were already aware of a possible problem at KC. The first mention of a trading systems’ problem, by a KC spokeswoman, appears at 10:01 AM. Based on the Bloomberg articles we review in the Internet Appendix, it seems clear that by 10:41 AM a wide range of active liquidity providers had become aware of the trading glitch that occurred at KC. Since information percolation among active liquidity providers is likely faster than what we can infer based solely on Bloomberg, it seems reasonable to assume that by 10:40 AM the trading glitch was common-knowledge among major traders, and in particular KC’s competitors. There may have been some remaining uncertainty regarding whether some of the glitch trades would be cancelled (some of the articles we review in the Internet Appendix suggest that by noon this uncertainty was resolved), but the possible cancellation of some early morning trades should not have prevented ongoing ‘risk-arbitrage’ activity to bring prices back in line with pre-glitch levels. And yet, as we show below the price impact of the trading glitch took until the end of the trading day to revert.

KC’s traders managed to reduce the size of the positions on the day of the glitch, but KC would not have been able to open for business on the next day because of regulatory capital requirements. As a result, KC struck a deal with Goldman Sachs Group Inc. to sell its portfolio at a 5% discount.⁹ While the glitch happened on Wednesday August 1, KC’s survival was still uncertain on the following Friday because of the large trading loss. In particular, KC secured a credit line to stay in business through late Friday and was actively involved with investors and rivals to secure additional funding.¹⁰ On the next Monday morning, it was announced that KC had signed a \$400 million rescue package from six other firms. In December 2012, KC was acquired by GETCO, a competing firm.

The short-term survival of KC was an important issue since KC was a designated market maker (DMM) on the NYSE. DMMs succeeded NYSE specialists. They have “obligations to maintain a fair and orderly market in their stocks, quote at the NBBO a specified percentage

⁸SEC Administrative Proceeding File No. 3-15570 (October 16, 2013) available at <https://www.sec.gov/litigation/admin/2013/34-70694.pdf>.

⁹Source: “Knight Held \$7 Billion of Stocks Due to Glitch,” *Wall Street Journal*, August 13, 2012.

¹⁰Source: “Knight Capital Group hustling to find a buyer or secure funding,” *Los Angeles Times*, August 4, 2012.

of the time, and facilitate price discovery throughout the day as well as at the open, close and in periods of significant imbalances and high volatility” (NYSE fact sheet, 2009).

The trading glitch impaired KC’s ability to act as a DMM. First, KC’s market making activity appears to have dropped substantially following the glitch—in part because market participants such as retail brokers stopped routing their orders to KC. Second, NYSE reassigned stocks for which KC was the DMM to GETCO on August 6, 2012.¹¹

2.2. The Impact of the Glitch on the Level of Prices

[Insert Figure 1 here]

The glitch resulted in large price movements and high volume in affected stocks. Several stocks had larger share volume than SPY—the most traded S&P 500 ETF—during the initial 30 minutes of trading on August 1. Panel (a) of Figure 1 shows that stocks that were bought as part of the glitch experience an average positive jump of about 4% concurrent with the heavy glitch-induced trading volume.¹² About half of the jump reverts by 10:30 AM when news of the glitch has become public, but it takes until market close before prices revert to their pre-glitch levels. The picture of the price-path followed by stocks sold as part of the glitch is the mirror image of that of bought stocks: they initially drop by about 4%, recover half of the drop by 10:30 AM, and revert fully by 4:00 PM. Interestingly, we find that both groups of stock over-shoot their ‘equilibrium’ level around 3:00 PM, which correlates with a significant increase in trading.¹³

Panel (b) shows that the average share volume of sold (bought) glitch-affected stocks are about 10 (7) times higher than non-glitch affected stocks during the glitch. Then shortly after 10:00 AM, the volume on the glitch-affected stock drops considerably. For stocks that were bought as part of the glitch the volume remains about 8% higher than normal at the end of the day. We also observe that around 3:00 PM the volume is abnormally high for

¹¹Sources: “Knight Capital heads into make-or-break weekend,” *Reuters*, August 4, 2012; “NYSE Euronext temporarily re-assigns Knight Capital DMM responsibilities to GETCO DMM for certain securities,” *Business Wire*, August 6, 2012.

¹²We describe how we identify stocks affected by the glitch in Section 4.3. Among glitch-affected stocks, a stock is identified as bought (sold) if it has a return greater or equal than 1% (lower or equal than -1%) over the first 30 minutes of trading on August 1. The results are similar if we use thresholds of 2.5% and -2.5%.

¹³At 2:55pm, the overshooting for sold stocks is statistically significant with a t-statistic of 3.56. The overshooting for bought stocks is significant at the level of 10% with a t-statistic of -1.84. One caveat regarding the statistical significance of the overshooting is that we did not fix the 3pm time ex-ante, but rather observe it ex-post. Given the volatility of daily fluctuations in these stocks, see figure IA.1 in the internet appendix, it is perhaps not surprising to find *some* time of day where prices deviate significantly from zero. That said, it is striking that at that *same* time bought and sold stocks deviate significantly in opposite directions.

glitch-affected stocks, which coincides with the period when stocks over-shoot their opening price.

The price patterns of glitch-affected stocks are consistent with slow-moving capital, where it takes a day for the market to absorb a (positive or negative) unexpected supply shock, even though it is unrelated to fundamental news. It is plausible that early on there was some uncertainty about the motive for the heavy buying and selling. So an increase (decrease) in price due to heavy buying (selling) can be initially ascribed to the market participants worrying about informationally-motivated trades. By mid-morning, however, the glitch was public news (see Section 2.1). Also, in contrast to what occurred around the ‘flash-crash’ for example, there was no contemporaneous abnormal move in the VIX during this episode, which suggests that markets did not experience a market wide change in ‘risk-aversion.’ Therefore, in the absence of frictions, one would expect the price to revert, on average, to the pre-glitch price as soon as it is common-knowledge that the heavy trading was not informationally motivated. Instead, it takes the whole day for the price to revert to the pre-glitch price.

[Insert Figure 2 here]

In Figure 2, we show the price pattern one would expect to obtain in four different models. First, we plot the price one would expect to see in a standard ‘neo-classical’ model where a positive (left panel) or negative (right panel) supply shock is observed in period 1 and investors are uncertain about whether it is informationally motivated. The price rises for bought stocks and decreases for sold stocks. However, as soon as investors learn (at the end of period 1) that the shock is not informational, the price should rebound to its pre-shock value. Instead, if liquidity providers have limited risk-bearing capacity (e.g., because they are risk-averse as in Grossman and Miller (1988)) then, because of inventory risk considerations, we expect the price pattern plotted as a solid line in Figure 2. Specifically, stocks that experience a positive (negative) supply shock decrease (increase) and then subsequently revert at the same rate as the supply shock reverts back to zero, so that liquidity providers are adequately compensated for the inventory risk they incur. The pictures are generated by a standard rational expectation model of the type developed in Duffie (2010) with competitive risk-averse market makers that respond to an unexpected mean-reverting supply shock that occurs in period 1. We present the model and its solution in Appendix A.

In addition, if we assume that some of the traders only trade infrequently (as in the ‘slow-moving capital’ model of Duffie (2010)), then the model can also generate the overshooting that we observe in the data around 3 PM. This is illustrated in the figure by the dashed line, which plots the price path corresponding to the ‘slow-moving capital’ model, presented

in Appendix A. Adding infrequent traders has three consequences. First, the impact of the supply shock is exacerbated at date 1. This is because the same unexpected supply shock must be accommodated by a smaller number of traders (as some of the infrequent traders are not present in the market at the time of the shock). Second, as the original positive (negative) supply shock reverts back to zero, the effective supply, which corresponds to the supply net of the holdings of the infrequent traders who absorbed some of the original supply shock and now stay out of the market, will actually become negative (positive)—in the figure around period 4. This leads the price reversal pattern to overshoot the fundamental value. Lastly, when the infrequent traders who absorbed some of the original supply shock come back to the market, they effectively propagate the initial supply shock, which generates a periodic positive autocorrelation in shocks. This creates ‘wave’ like patterns in stock prices (see Duffie (2010) and Bogousslavsky (2016) for further discussion). Thus, the ‘slow-moving’ capital model provides one possible explanation for the ‘over-shooting’ pattern that one observes in data (and for the symmetric pattern observed for bought versus sold stocks).

On the other hand, in the model prices eventually revert to their pre-glitch level when the supply has reverted to its pre-glitch level. In the data, we know that at the end of the trading day, KC retained a large portfolio of the glitch stocks that it had not been able to offload in the stock market. This seems somewhat at odds with the idea that the initial supply shock had fully reverted back to pre-glitch levels. Further, it is not entirely obvious that the glitch is isomorphic to a supply shock as in the standard model. For example, one might conjecture that strategic considerations, both on the part of KC to mitigate its trading costs and on the part of the market to extract rents from its knowledge of KC’s desired trades, would affect the price paths of glitch-affected stocks. While we cannot entirely rule out that strategic considerations are at play, the overall pattern does not seem to be consistent with such effects. Indeed, we would expect strategic traders with strong market power to buy (sell) back stocks from (to) KC only at better than pre-glitch prices. So, in a predatory trading model with highly strategic traders, who know the ‘fundamental’ pre-glitch value of the stock and that KC has to trade out of its position, we would expect prices of stocks that were bought (sold) as part of the glitch to immediately revert below (above) their pre-glitch price as illustrated by the dash-dotted line in Figure 2. We do not observe such pattern, but instead a gradual reversion of prices during the day to their pre-glitch level, with only a short period of over-shooting around 3 PM.

Qualitatively, the price level evidence is consistent with the price drop and reversal documented by Coval and Stafford (2007) and Edmans et al. (2012) around mutual fund fire sales. However, quantitatively for a similar magnitude return drop the fire-sales literature

finds a much longer reversal time of close to two-years.¹⁴ Taken at face value these papers suggest very slow-moving capital. In his discussion, Duffie (2010, p. 1260) suggests that this slow reversal might be the consequence of the difficulty in markets to separate a pure supply shock from an informationally-motivated trade. Duffie (2010) conjectures it might take the market several months to learn that the fire-sales, identified as such ex-post by researchers, were not, in fact, informed trades. In a recent paper, Wardlaw (2019) also finds that the pure price pressure effect may be a lot shorter and less important than measured in these previous papers. Consistent with this view, our results suggest that if the market quickly learns that the trade is not informationally motivated, then the reversal is a lot quicker, i.e., an average 4% drop due to glitch-sales reverts in one day (instead of 24 months). Furthermore, we find that the effect is symmetric for glitch-buys.¹⁵

Having established that the impact of the unexpected (negative and positive) supply shocks on the level of prices is significant and takes one full day to revert, we investigate how stock liquidity is affected by the glitch.

2.3. Potential Impact of the Glitch on Liquidity

The KC trading glitch provides a natural experiment to evaluate specific theories of stock market liquidity. Indeed, liquidity could be affected by the glitch for several reasons.

First, the trading glitch represents a large shock to the inventory of a market participant—KC held aggregate long and short positions worth billions of dollars. If KC has limited risk-bearing capacity, it will seek to unload its positions. We would expect trading costs to increase for affected stocks in the following hours and days as the supply shock propagates across investors trading at different speeds. An increase in KC's trading intensity to unload positions should increase trading costs (e.g., Barra, 1997). Importantly, we conjecture that capital market imperfections such as slow-moving capital and predatory trading are likely to accentuate the increase in trading costs.¹⁶ In our empirical tests, we control for well-known determinants of liquidity such as volume and volatility to focus on the excess cost of

¹⁴Coval and Stafford (2007) find a 15% cumulative return drop due fire sales that starts 12 months before it begins to revert and takes more than 18 month to revert (see their figure 2). Edmans et al. (2012) find a 5% return drop around large unexpected fire-sale orders that takes more than 24 months to revert (see their Figure 2).

¹⁵We report in the Internet Appendix the price patterns of bought and sold stocks over the next five trading days. The small number of stocks in each group (14 and 17) requires caution when interpreting the patterns over several days as idiosyncratic events are likely to occur. The evidence suggests that the reversal is permanent.

¹⁶The evidence in Section 2.2 is more supportive of slow-moving capital than predatory trading. We note, however, that more refined versions of the slow-moving capital model of Duffie (2010) and the predatory trading model of Brunnermeier and Pedersen (2005) are needed to study the dynamics of trading costs following a supply shock.

trading following a (very) large inventory shock. Interestingly, in the KC case it is known that Goldman Sachs bought KC's portfolio at a discount on the day of the glitch.¹⁷ Clearly, Goldman Sachs still had to unload the position due to its balance sheet constraints, thus propagating the supply shock further. We label this theory the 'supply shock channel.'

Second, the trading glitch represents an exogenous shock to a market maker's capital. Less capital should limit the ability of a market maker to provide liquidity (Grossman and Miller, 1988). Comerton-Forde et al. (2010) show that NYSE specialist inventories and trading revenues generated by overnight inventories predict bid-ask spreads for the next day. Relatedly, Coughenour and Saad (2004) document the existence of a common component in liquidity (as measured by spreads) for all stocks assigned to a particular specialist. These papers do not rely on an exogenous shock to identify the causal relation between market maker's capital and stock liquidity. In addition, these papers cover periods prior to the switch to designated market makers on the NYSE. We label this the 'market making capital channel.'

These two channels makes several predictions that may be complementary. First, they both predict that the shock should lead to an increase in trading costs. Second, these trading costs should take some time to revert to normal. The speed of mean-reversion provides an estimate of the time it takes for the inventory risk to be spread-out across investors or equivalently for market-making capital to be reallocated.

However, they also make specific predictions. The supply shock channel predicts an asymmetric impact on stock market liquidity: it should be especially costly to trade in the opposite direction of the inventory imbalance. For instance, if KC has accumulated a large long position in a stock, it should be more costly to sell than to buy the stock in the aftermath of the glitch. That is, trades that supply liquidity to KC by taking off some of the excess inventory should be cheaper than trades in the opposite direction. The market making capital channel predicts that stocks for which KC is the designated market maker but that are not affected by the glitch should also experience a decrease in liquidity.

The fact that KC had off-loaded its position to GS on the day of the glitch is particularly helpful to isolate the effect of the 'supply-shock channel.' Indeed, if we observe a persistent effect that is larger for stocks that experienced the glitch compared to those that did not, this can hardly be ascribed to the market-making capital channel, since KC did not hold these large inventory positions any longer.

The supply shock channel predicts that turnover in the glitch-affected stocks should remain elevated as long as it takes for the excess supply of stocks to be reallocated from fast

¹⁷The news appeared in the press two days after the glitch, on Friday afternoon (see, for instance, "Goldman support boosts Knight Capital," *Financial Times*, August 3, 2012).

traders to slow traders. Instead, the market making capital channel predicts that because of a lower risk-bearing capacity of liquidity providers there should be less turnover (in other stocks) as the market maker is less willing to supply liquidity.

We test these predictions in our analysis.

3. Liquidity Measures

In this section, we describe the liquidity measures that we use in the analysis with an emphasis on the two most frequently used cost measures for institutional trading: implementation shortfall and VWAP slippage.

3.1. Implementation Shortfall

As a first measure of institutional trading costs, we define implementation shortfall (IS) as introduced by [Perold \(1988\)](#). IS is the widely-preferred measure of trading cost for institutions and has been frequently employed in the literature to proxy institutional trading cost. IS is based on comparing the weighted average price of the actual trades to a benchmark price observed prior to the parent-order execution period. Formally, the IS of a parent-order execution k is given by

$$(1) \quad \text{Implementation Shortfall}_k = \text{sgn}(Q_k) \frac{P_k^{\text{avg}} - P_{k,0}}{P_{k,0}},$$

where P_k^{avg} is the value-weighted execution price of the parent-order execution and $P_{k,0}$ is the mid-quote price of the security at the beginning of the execution period.

3.2. VWAP Slippage

Implementation shortfall considers an ex ante static benchmark and does not consider any permanent price changes during the parent-order execution period. For this reason, one popular ex-post benchmark for average execution price is the volume-weighted average price (VWAP) during the trading interval. Formally, the VWAP in a particular trading interval period is defined as follows. Suppose that there are N trades during this period. Given the sequence of trades at prices $P_1, \dots, P_N$ with corresponding quantities $V_1, \dots, V_N$, the VWAP is given by

$$(2) \quad \bar{P} = \frac{\sum_{i=1}^N P_i V_i}{\sum_{i=1}^N V_i}.$$

Using this definition, the VWAP Slippage for a parent-order execution k is given by

$$(3) \quad \text{VWAP slippage}_k = \text{sgn}(Q_k) \frac{P_k^{\text{avg}} - \bar{P}_k}{\bar{P}_k},$$

where $\bar{P}_k$ is the realized VWAP over the parent-order execution period.

3.3. Standard Liquidity Measures (Spreads, Depth, and Turnover)

We compute the following spread measures for a transaction on stock i at time t :

$$(4) \quad \text{Effective Spread}_{i,t} = 2|\ln P_{i,t} - \ln M_{i,t}|,$$

$$(5) \quad \text{Realized Spread}_{i,t} = 2q_{i,t}(\ln P_{i,t} - \ln M_{i,t+5\text{mn}}), \quad \text{and}$$

$$(6) \quad \text{Price Impact}_{i,t} = 2q_{i,t}(\ln M_{i,t+5\text{mn}} - \ln M_{i,t}),$$

where $M_{i,t}$ denotes the midpoint of the best quote available immediately preceding the transaction, and $q_{i,t}$ equals 1 for buy orders and -1 for sell orders.¹⁸ The average spread over any interval is then computed by summing the weighted spread associated with each transaction over the interval, where the weight equals the dollar volume of the transaction over the total dollar volume in the interval. We follow the literature and compute realized spread and price impact using the quote midpoint five minutes after a trade.

Another important standard measure of liquidity is depth. Average total daily (dollar) depth is computed as the sum of the time-weighted depth at the best bid and the time-weighted depth at the best ask.

Finally, we use share turnover as another measure of liquidity. A higher turnover is generally associated with a higher liquidity in the cross-section. Furthermore, an examination of turnover may indicate whether abnormal patterns in trading activity arise following the glitch.

4. Data and Methodology

4.1. Data

We use proprietary large-order execution data from the historical order database of a large investment bank providing algorithmic trading services (“The Bank”). The Bank is one of the top five banks in the United States with respect to total assets and is one of the top five

¹⁸We focus our analysis on percentage spreads. We obtain similar results with dollar spreads.

providers of execution services by market share.

This dataset has been originally obtained in May 2013 to study the implications of investor heterogeneity on the estimation of the price impact. For this purpose, the executions in the sample have been selected from an active subset of the broker's clients and the liquid subset of the stock universe, S&P 500 stocks. This data set consists of 81,744 parent-order executions submitted by 293 unique investors between January 2011 and December 2012. In Appendix B, we provide detailed information about how the data set is constructed and provided by the Bank.

The large-order executions originate from a diverse pool of investors, such as institutional portfolio managers, quantitative investment funds, internal trading desks and retail brokers. The data set consists of two frequently used execution algorithms in the industry, the volume weighted average price (VWAP) and the percentage of volume (POV). The VWAP algorithm is designed to achieve an average execution price that is as close as possible to the volume weighted average price over the trading interval. Similarly, the main objective of the POV algorithm is to have constant participation rate in the market within the trading interval. These two algorithms are conceptually similar but they specifically differ in pre-trade instructions submitted by the investor. In a VWAP order, the investor specifies his urgency in execution by specifying a target completion time. The schedule is then determined using the historical volume curve between the initiation and target completion. In a POV order, the investor specifies minimum and maximum bounds on the desired participation rate level which ultimately determines the urgency of the trade.

The data set provides a rich set of attributes. For each parent-order execution, the data contain parent-order- and stock-level statistics. Parent-order-level statistics include the number of shares to be executed, direction of the execution (buy or sell), execution start and end times, participation rate (the ratio of execution size to the total volume during the trading interval), average execution price, proportional bid-ask spread and mid-quote volatility based on the duration of the execution. Stock-level information includes average daily volume, proportional bid-offer spread and mid-volatility of the stock on the trading day along with their rolling averages over the last 20 trading days prior to the execution. As mentioned earlier, we also have the masked identity of the investor submitting the order.

We filter the original data set to obtain an appropriate data to investigate the slow-moving capital theories around the exogenous glitch event. First, due to the event date, we exclude all of the parent-order executions prior to January 2012. This removes 42,571 parent-order executions. Second, we restrict our sample to NYSE-listed stocks since all of the stocks affected by the glitch are NYSE-listed. Furthermore, the NASDAQ does not have designated market makers. Hence, including NASDAQ stocks may potentially bias our results because

of the different market structures. This removes 7,381 parent-order executions.¹⁹

After these restrictions, our sample consists of 31,792 parent-order executions on 394 S&P 500 stocks which are traded in 2012 and submitted by 266 unique clients. Parent-order executions consist of 14,687 buy and 17,105 sell executions. There are 23,194 VWAP and 8,598 POV parent-order executions, respectively. For 2,746 VWAP parent-orders that occur in the first quarter of 2012, we have the complete child-order data set.²⁰ Due to its time coverage, the child-order data set is not applicable for the main analysis regarding the glitch period, but to provide more information about a typical parent-order execution, we have computed that the average (median) number of child-order trades in a parent-order is 113.9 (62). Thus, this statistic implies that the child-order data corresponding to our parent-order data set that covers 2012 would have more than 3.5 million individual trades. Overall, these statistics are useful to compute a back-of-the-envelope estimate for the number of individual child-order trades in a typical parent-order.

Using the available child-order fills, we also compute the fraction of aggressive, passive and mid-point fills. We define a buy (sell) child-order trade as an aggressive fill if transaction price occurs at a price higher (lower) than the mid-point of the National Best Bid and Offer (NBBO) realized during the same second. Similarly, passive fills correspond to child order executions if the transaction price is lower (higher) than the NBBO mid-point when the investor has a large buy (sell) order. Finally, mid-point fills are transactions happening at the mid-point of the NBBO. We then create the corresponding parent-order level statistics, by computing the fraction of the shares executed via each order type. We find that on average, 59.0% (31.5%) of the parent-order execution is filled aggressively (passively). We expect these ratios to go up for the POV algorithm due to its higher participation rate, on average. Overall, these statistics imply that the algorithms in our data set frequently utilize market orders for child order execution.

Panel (a) of Table 1 reports additional summary statistics about our parent-order executions.

[Insert Table 1 here]

First, parent-order executions are sizable with average (median) order value of \$0.92 (0.34) million. These values correspond to a wide range of participation rates, the ratio of

¹⁹In the robustness checks, we include NASDAQ S&P 500 stocks and obtain similar results.

²⁰Each parent-order consists of several child-orders, which can be routed to public exchanges and dark pools for execution. We have access to the complete child-order data set for a subset of the parent-orders which are VWAP orders and occur between January 2012 and March 2012. For this portion of the dataset, we know the time (timestamped to the millisecond), size, venue (market center) and price of each child-order trade.

the order size to the total volume, with an average (median) ratio of 5.34% (0.95%). Average and median implementation shortfall are 3 bps. Average parent-order execution duration is 2.65 hours. Finally, average and median percentage return realized during a parent-order execution are roughly zero. Finally, the data set does not concentrate on any single stock. The highest number of parent-order executions on a single stock is 327 (NYSE:WFC) which corresponds to only 1.03% of all executions.

Panel (b) of Table 1 provides the summary statistics for parent-order executions that occurred on August 1, 2012. There are 662 parent orders executed by The Bank on that day. The maximum number of parent-order executions in a single stock from the affected group is 3 (SWN and WFC) resulting in a homogeneous distribution between the affected stocks. More than 90% of the parent-order executions start after the end of the glitch, specifically, after 10:30 AM. Therefore, our data mainly captures changes in liquidity in the aftermath of the trading glitch as opposed to during the glitch.

We complement our parent-order execution data set by obtaining stock information from the Center for Research in Security Prices (CRSP), earnings announcement dates from I/B/E/S, and intraday data from the Trade and Quote (TAQ) database (Monthly TAQ). Spreads and depths as described in Section 3.3 are computed using the methodology and filters detailed in Holden and Jacobsen (2014). Trades are signed with the Lee and Ready (1991) algorithm using the first quote available prior to a trade.

Table 2 provides descriptive statistics for the standard liquidity measures as well as various stock characteristics for the stocks in our sample in 2012. The average (median) effective spread is 4.41 (3.42) basis points, which is small. This is not surprising given our focus on NYSE-listed S&P 500 stocks. The median daily turnover is 0.75% and the median market capitalization is 12.42 billion dollars.

[Insert Table 2 here]

4.2. Comparison of the Data with other Institutional Trading Data

In terms of average order size (0.92 million) and participation rate (5.34%), our institutional trading dataset is very similar to other datasets studied in this literature. In this literature, the most commonly used dataset is provided by Abel Noser (formerly ANcerno) which is a consulting firm providing transaction cost analysis to institutional investors. Puckett and Yan (2011) study this dataset between 1999 and 2005 and report that the trading data accounts for approximately 8% of the dollar value of CRSP trading volume. Using an extended sample through 2008 but with further filters to identify the broker that executes the trade, Anand et al. (2011) report that the average daily participation of the institutions

is 2.1% between 1999 and 2008. This average drops to 1.0% in 2008. Further, they report that the average number of child-order trades is only 2.13 as opposed to 113.9 in our dataset.

Hu et al. (2017) find that the Abel Noser dataset may account for up to 12.3% of the total CRSP dollar volume between January 1999 and September 2011. Hu et al. (2017) also acknowledge the declining trend in Abel Noser’s volume coverage and state that the coverage drops to 6.4% in 2011. Further, they find that institutional trade sizes drop dramatically over time. Specifically, they report that the median dollar trade size in the dataset is only \$4,708 in 2011 compared to \$342,533 in our dataset. Last, they report that the Abel Noser dataset may contain unreliable intraday timestamps, which does not allow the researcher to know the exact start- and end-time of the parent-order execution (i.e., the order duration).²¹ Nevertheless, in Section 7.6, we utilize a small sample from this data set around the glitch period (after excluding the small orders) and find that our main findings can be replicated in this sample as well.

Korajczyk and Murphy (2018) and van Kervel and Menkveld (2017) employ institutional trading data from Canada and Sweden to examine the impact of high-frequency trading on execution costs. Both of these datasets do not exactly identify how the parent-orders are split into child-orders and thus, they infer the parent-order by constructing sequences of pure-buy or pure-sell trades belonging to the same investor. In our dataset, parent-order level statistics are exactly reported. Korajczyk and Murphy (2018) report an average trade size of \$2.2 million with a participation rate of 2.5%. van Kervel and Menkveld (2017) examine roughly 5,000 orders from Sweden during a similar period to ours and report an average trade size of \$2.2 million corresponding to a participation rate of 3.6%. Overall, these similar statistics in order size and participation rate support the representativeness of our institutional trading data.

4.3. Identification of Affected Stocks

Available public data do not allow us to identify stocks affected by the trading glitch with certainty. However, according to a KC statement on August 2, the erroneous orders were limited to NYSE-listed stocks. Further, we observe highly abnormal trading volume for several NYSE-listed stocks in the first half-hour of trading on August 1 absent any specific corporate news. We discuss below how we use this abnormal increase in volume to identify the stocks affected by the glitch.

Given that the number of trades drops significantly after 10:00 AM, we focus on the number of trades occurring in the first half-hour of trading. As a benchmark, the number

²¹Finally, this dataset does not report the client identifiers after September 2011 and is currently unavailable for purchase by academic researchers.

of trades for the most active S&P 500 ETF, SPY, is 31,341. Our first identification criterion is simply to classify as glitch-affected each stock with a higher number of trades in the first half-hour of trading than this benchmark number. We verify that this condition is extremely rare by analyzing the prior year period from August 2011 to July 2012. Out of approximately 100,000 stock-day observations in this period, there are only 36 events in which a stock has had higher number of trades than that of SPY in the first half-hour of trading.²² On August 1, we find that 35 stocks have more trades than SPY and label these stocks as *glitch-affected*.²³ In Table IA.1 of the Internet Appendix, we report historical statistics of each affected stock's number of trades compared to that of SPY within the first half-hour of trading. We find that 31 stocks have never had a trading day with higher number of trades than that of SPY during the period between August 2011 and July 2012. Typically, the average ratio of the number of trades of the affected stock to that of SPY is less than 10%. These statistics strongly suggest that the trading levels on these stocks are unusual.

In addition, we compute for each stock the average number of trades between 9:30 and 10:00 AM over the past five trading days. We label a security as glitch-affected if the ratio of its number of trades on August 1 (between 9:30 and 10:00 AM) over its average number of trades over the past five trading days (between 9:30 and 10:00 AM) is larger than 10. This condition adds another two stocks.²⁴ In total, we label a total of 37 stocks as glitch-affected.²⁵ In Section 7.4, we show that our findings are robust to a different selection criterion of glitch-affected stocks. Instead of benchmarking against SPY, which may oversample large firms, we use stock-level historical statistics on the number of trades realized in the first half-hour trading to identify stocks affected by the glitch. In addition, in Section 2 of the Internet Appendix we also use a list of potentially affected stocks provided by the NYSE at the time of the event as another robustness test.²⁶ In both analyses, our findings are very similar.

Panel (a) of Table 3 reports the list of glitch-affected stocks with their number of trades

²²10 stocks from the financial industry are involved in these 36 cases. The exact stock-event break-down is as follows: BAC has 17, C has 6, CHK and JPM have 3, RF has 2 and AIG, BBY, EMC, F and HPQ have one observation.

²³We exclude KC's own stock, KCG, from our analysis because it does not appear to be subject to a similar anomalous volume pattern. Trading volume increases only around 9:45 AM and is associated with a price decrease. This supports our conclusion from Section 2.1 that market participants quickly realized that KC was behind the abnormal volume and that this volume was not informed.

²⁴In Table IA.2 of the Internet Appendix, we report simple distributional statistics of this ratio over the prior year. We find that the mean and the median values of this ratio across large (top quintile) and small firms (bottom) are very similar whereas the variability is higher for small firms. The 99th percentile of the ratio for large (small) firms is 3.33 (4.59), which is considerably lower than 10.

²⁵This number is lower than that reported in the SEC's litigation release since we focus on NYSE-listed S&P 500 stocks.

²⁶As we show in Section 2 of the Internet Appendix, all but one stock that we identify using our volume based criteria are on the list provided by the NYSE. The NYSE list has a few additional stocks that did not, however, experience abnormal volume on the event day.

in the first thirty minutes of trading on August 1st. For each stock, we report the number of trades in the first half-hour of trading, the ratio of this number to the average number of trades over the past five trading days and the corresponding z-score of this number (using mean and standard deviation computed over the prior year). We find that 36 out of 37 stocks have z-scores of 8 or higher implying an extreme increase in the number of trades. In our institutional trading data, 32 out of 37 affected stocks are executed in 52 different parent-order executions on the day of the glitch.

[Insert Table 3 here]

Panel (b) of Table 3 reports the allocation of stocks among those affected by the glitch and those that have KC as DMM.²⁷ In what follows we refer to these stocks as *DMM stocks*. Interestingly, there is only one glitch-affected stock which has KC as its DMM. This property of the sample is valuable since it allows us to test whether stocks that are not affected by the glitch and have KC as DMM experience an increase in trading costs.

4.4. Methodology

We run the following regression to understand the determinants of trading costs around the glitch:

$$\begin{aligned}
 C_{i,k} = & \delta_0 \text{GlitchDay0}_{i,k} + \delta_W \text{GlitchNextWeek}_{i,k} + \delta_M \text{GlitchAug9ToAug31}_{i,k} + \delta_P \text{GlitchPost}_{i,k} \\
 (7) \quad & + \gamma_0 \text{DMM0}_{i,k} + \gamma_W \text{DMMNextWeek}_{i,k} + \gamma_M \text{DMMAug9ToAug31}_{i,k} + \gamma_P \text{DMMPost}_{i,k} \\
 & + \alpha_i + \alpha_t + \sum_j \beta_j \text{Control}_{j,i,k} + \epsilon_{i,k},
 \end{aligned}$$

where $C_{i,k}$ is our measure of institutional trading costs (VWAP Slippage or Implementation Shortfall) for stock i on parent-order execution k . Institutional parent-order executions are not indexed by stock and day since there can be zero or multiple executions for a given stock on the same day. Finally, note that this specification is equivalent to the standard difference in differences model with stock and calendar day dummies.

The first set of variables allows us to test the supply shock channel. That is, $\text{GlitchDay0}_{i,k}$ equals one if parent-order execution k is made on a (glitch-)affected stock on the day of the trading glitch. Similarly, $\text{GlitchNextWeek}_{i,k}$ equals one if parent-order execution k is made

²⁷We have obtained the list of DMMs and their associated securities from Nanex, a financial data analytics firm. KC is the DMM for approximately 500 securities to which we can match CRSP data. These securities include domestic stocks, closed-end-funds, foreign common shares, and REITs. As reported in the table, 18 NYSE-listed S&P 500 stocks have KC as DMM.

on an affected stock within the next 5 business days after the trading glitch (i.e., before August 9th) and $GlitchAug9ToAug31_{i,k}$ equals one if parent-order execution k is made on an affected stock on or after August 9th and on or before August 31st. Finally, $GlitchPost_{i,k}$ equals one if parent-order execution k is made on an affected stock on any date after August 31st.

The second set of variables allows us to test the market making capital theory. That is, $DMM0_{i,k}$ equals one if parent-order execution k is made on a stock for which Knight Capital is the designated market maker on the day of the trading glitch. We define $DMMNextWeek_{i,k}$, $DMMAug9ToAug31_{i,k}$, and $DMMPost_{i,k}$ in a similar way for parent-order executions made within the next 5 business days after the trading glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively.

Since both supply shock and market making capital channels (Section 2.3) can be at play in the data, it is important to estimate (7) with both sets of variables in order to avoid potential misspecifications. To separate the market making capital channel from the supply shock channel, we require a DMM stock not to be affected by the glitch for $DMM0_{i,k}$ (and all other DMM indicators) to be set to one. This restriction has little effect on our analysis since only one stock is affected by the glitch and has KC as DMM (Table 3).

A broad literature finds that the main determinants of variation in institutional trading costs are relative order size, market capitalization, asset volatility, share turnover, and bid-ask spread (e.g., Domowitz et al., 2001). Based on this evidence, we use the following set of stock- and parent-order-level characteristics as control variables:

- **Order size.** The price impact of a trade increases with order size. We capture this with the participation rate (the ratio of order size to interval volume) and the fraction of daily volume (the ratio of order size to daily volume).
- **Share volume and turnover.** All else equal, higher volume and turnover may signal higher liquidity. We use relative daily volume (the ratio of daily volume to average daily volume over the past month) and interval turnover (the ratio of trading volume realized during the lifetime of the parent-order to the number of shares outstanding) to control for differences in the volume profiles associated with a parent-order execution.
- **Volatility.** In the presence of higher volatility, market makers have the incentive to widen the spread between their quotes, which leads to higher trading costs. We control for this channel by using interval volatility (mid-quote volatility during the lifetime of the parent-order) and daily volatility (mid-quote volatility during the trading day).
- **Execution duration.** A higher urgency in trading is likely to result in higher trading

costs. We use execution duration expressed as a fraction of total trading hours to control for differences in the urgency level of parent-order executions.

- **Stock return.** We use the absolute and raw values of the daily stock return to control for extreme contemporaneous price movements.
- **Earnings announcement days.** We include a control variable, *IsEarningsDay*, which takes a value of 1 if the executed stock has an earnings announcement on the parent-order execution day.
- **Type of algorithm.** VWAP and POV strategy may differ in their expected trading costs all else equal. A POV strategy is usually employed for larger orders and thus can be intrinsically costlier. We include a control variable, *IsVWAP*, which takes a value of one if the parent-order execution algorithm is aimed to minimize VWAP slippage.

In addition to the above set of control variables, we include stock and day fixed effects in Equation (7). Standard errors are clustered by day and stock to control for heteroskedasticity, contemporaneous correlation across stocks, and autocorrelation within each stock (see [Petersen \(2009\)](#)).

With respect to our set of standard liquidity measures (Section 3.3), we estimate Equation (7) using spreads measured over the same time intervals as that of the parent-order executions. More precisely, we compute dollar-weighted spreads using only transactions and quotes between the start time and end time of each parent-order execution. As a result, each execution is matched to a spread computed over the same time interval. Hence, the same set of control variables can be used for spreads and institutional cost measures, which allows for a side-by-side comparison of both sets of measures.²⁸

5. Impact of the Glitch on Liquidity

In this section, we test the supply shock and market making capital channels using institutional cost measures and standard liquidity measures.

5.1. Institutional Trading Costs

As mentioned earlier, we have 52 parent-order executions on glitch-affected stocks occurring on August 1, 2012. There are 10, 9, 6, 15 and 20 parent-order executions from the affected

²⁸One issue that arises with Equation (7) is that we do not have a complete day-stock panel as some stocks may be executed more than once in a day or may not be executed at all.

group in the next five business days following the glitch, respectively, totaling 60 parent-order executions. Furthermore, we have 215 and 1,031 affected parent-order executions in the remaining days of August and during the last quarter of 2012, respectively. We emphasize again that each parent-order execution is estimated to contain a median of 62 child-orders (Section 4).

We first examine visually whether there is any increase in the institutional trading costs of glitch-affected stocks. For this purpose, for each day, we compute the mean difference in VWAP slippage between the parent-order executions on the affected and unaffected groups of stocks during the period of July 2012 and September 2012 (inclusive).

Figure 3 illustrates the mean and the median of the slippage differences across different horizons around the glitch period. For both mean and the median statistics, the difference in slippage spikes during the glitch day and the week following it. We observe that the mean and the median of the difference distribution drop during the post-glitch period. The mean of the difference in September becomes roughly the same as the mean realized during pre-glitch period.

[Insert Figure 3 here]

Next, we estimate equation (7) using VWAP Slippage and Implementation Shortfall (IS) as cost measures. Table 4 reports the regression estimates.

[Insert Table 4 here]

Participation rate, fraction of daily volume, execution duration, and the type of the trading strategy (VWAP or POV) are the statistically significant variables for both cost metrics. As expected, execution costs are positively correlated with participation rate, fraction of daily volume, and execution duration. Our cost model explains a higher variation of VWAP Slippage than IS.

The main variables of interest are the indicators associated with parent-order executions on glitch-affected stocks during and after the trading glitch. The results confirm the visual evidence in Figure 3. Institutional trading costs are economically and statistically significantly higher on the day of the trading glitch for affected stocks. The estimated cost increase is approximately 6.4 and 18.2 basis points for VWAP Slippage and IS, respectively. This estimated cost increase is economically substantial since the median VWAP Slippage (IS) in our data set is 1.0 bp (3.1 bp).

We can also interpret the estimated coefficient by comparing it with the most important parent-order-level parameter, the participation rate. A 6.4 (18.2) bps cost increase in VWAP Slippage (IS) corresponds roughly to an additional 45% (50%) increase in participation rate.

That is, all else equal, a glitch-affected parent-order execution with a participation rate of 10% is expected to have a VWAP Slippage (IS) similar to that of an unaffected parent-order execution with a participation rate of 55% (60%).

Table 4 formally shows that institutional trading costs remain higher for more than one week after the glitch. Strikingly, parent-order executions on glitch-affected stocks occurring within the next 5 business days are exposed to roughly the same cost increase as on the day of the glitch for both cost measures. As measured by VWAP slippage (IS), parent-order executions in the affected group cost on average an additional 6.5 (16.7) bps in the following week.

Furthermore, VWAP Slippage implies an even more persistent cost increase. The indicator variable, *GlitchAug9ToAug31*, is significant for VWAP Slippage and indicates an additional 1.1 bp cost increase for the rest of August. Although this increase is roughly 17% of our earlier estimates of the cost increase, it remains a substantial cost considering the additional three-week horizon and that the median VWAP Slippage is merely 1 bp.

Using these estimates for the cost increase, we make a simple back of the envelope calculation to get a sense for the (order of) magnitude of the total loss incurred by liquidity-seeking institutional investors (assuming that our executions are representative of average institutional executions).²⁹ In July 2012, the average dollar volume per day on the glitch-affected set of stocks was roughly 20 billion. Given that a majority of the daily trades are due to institutional investors, we attribute 10 billion of dollar volume to institutional investors and whose trades are intermediated by fast HFT liquidity providers. Thus, using IS (VWAP Slippage) as our cost measure, the total loss to the institutional investors due to the trading glitch is approximately \$16.7 (\$6.5) million per day during the first five days. Thus, the total IS loss for the first five days amounts to \$83.5 million. This is admittedly a rough estimate since it presumes that the increase in transaction costs observed around the glitch entirely results in a transfer from institutional investors (e.g., pension and mutual funds, insurance companies) to high-frequency traders. In practice, this transfer could be smaller if some of the volume occurred without going through intermediaries. Brogaard et al. (2014) find that high frequency traders are involved in roughly 42% of trading volume in a sample of large stocks over 2008-9. In this case, our institutional loss estimate should be divided by a factor between two and three.

In summary, Table 4 shows that the glitch has a large impact on the liquidity of affected stocks as measured by institutional trading costs. The increase in trading costs persists for several days after the glitch. That is the effect on liquidity endures even after KC has

²⁹Clearly there are many ad-hoc assumption that go with this back of the envelope calculation, which should be taken simply as an illustration of the potential implications of our estimates.

sold off its portfolio to GS and even after its DMM responsibilities have been reassigned to Getco. This evidence strongly supports the supply shock channel described in Section 2.3. Importantly, institutional trading costs increase despite controlling for the participation rate (volume) and volatility. This suggests that the increase in trading costs goes beyond a standard volume-price impact explanation and points to the role of market frictions such as slow-moving capital (Section 2.2). For a given volume and volatility, the cost of trading glitch-affected stocks is higher in the aftermath of the glitch.

5.1.1. Market Making Capital Channel

Next, we examine the DMM coefficients in (7) to test the market making capital channel. These coefficients identify the impact on institutional cost measures for parent-order executions that are on stocks that have KC as DMM and are *not* affected by the glitch. We have 14 parent-order executions occurring on August 1, 2012 and we have an additional 20 parent-order executions during the next 5 trading days.

Table 4 shows that institutional trading costs are significantly higher, both economically and statistically, for DMM stocks on the day of the glitch. This result is consistent with the market making capital channel: stocks that are not directly affected by the glitch experience an increase in trading costs because the designated market maker for these stocks is limited in its ability to provide liquidity due to a large capital loss. The natural experiment provided by the trading glitch supports this channel since it is difficult to lay out alternative explanations as to why the cost of trading these stocks would increase otherwise.

Table 4 does not, however, indicate significant changes in the post-glitch period for DMM stocks. First, as noted above we only have few parent-order executions on DMM stocks in the post-glitch period, which limits the power of our test. Second, KC's market making responsibilities were transferred to GETCO on August 3, 2012, which may in theory explain why the effect is short-lived. We present additional tests of the market making capital channel below using spreads on an extended sample of stocks.³⁰

As explained in Section 2, KC sold its portfolio at a discount to Goldman Sachs, which was also a DMM. Since Goldman had to absorb a large inventory, its ability to act as DMM might have been impaired. Contrary to KC, Goldman suffered no capital loss. To assess this effect, we include in (7) dummies for executions made on the day of and in the aftermath of the glitch on stocks for which Goldman is a DMM. First, all of our previous results are robust to including the Goldman DMM indicators.³¹ Second, we find weak statistical evidence that

³⁰We extend the sample to include non-S&P 500 stocks for which we can compute spread data but for which we do not have institutional parent-order execution data.

³¹The majority of glitch-affected stocks do not have Goldman as DMM, which rules out an indirect market

implementation shortfall increases by approximately 11 bps on these stocks in the week after the glitch. There is a small and statistically insignificant increase in VWAP slippage.

5.1.2. Asymmetric Impact of Supply Shock on Liquidity

Institutional trading costs on glitch-affected stocks increase persistently in the aftermath of the glitch, consistent with the supply shock channel discussed in Section 2.3. The supply shock channel further predicts an asymmetric impact of inventory on liquidity. Following a supply shock, it should be especially costly to trade in the opposite direction of the shock.

We test this prediction using large positive or negative returns realized during the trading glitch to proxy for changes in inventory. If a stock had a large positive (negative) return during the glitch, then KC is likely to have built a large long (short) position on the stock. More precisely then, we test whether institutional orders that execute in the same direction as KC's hypothetical liquidation orders following the glitch cost more for stocks with extreme returns during the glitch period. We create a dummy variable, *KnightSameSide*, that equals 1 if the stock realized a $\pm 2.5\%$ return during the glitch period and if the institutional investor trades in the opposite direction of the extreme return; that is, in the same direction as KC if they liquidate the accumulated inventory. We consider all the parent-order executions occurring between August 1st and August 8th and find 21 parent-order executions that qualify as *KnightSameSide*=1.³²

Table 5 shows that these parent-order executions incur significant additional costs for both VWAP Slippage and Implementation Shortfall metrics. These parent-order executions cost an additional 9 (31) bps in VWAP Slippage (IS), which is economically significant.

[Insert Table 5 here]

5.2. Spreads

Liquidity measured by institutional trading costs decreases around the glitch for glitch-affected stocks and DMM stocks. We now compare this evidence to that provided by spreads matched to the parent-order executions. The spreads are computed over the time interval of the parent-order executions so that the measures can be exactly compared and the same set of control variables can be used (see Section 4.4).

Columns 3-5 in Table 4 report the results. Matched effective spreads on glitch-affected stocks increase on the day of the glitch by 0.86 bp on average. This increase is statistically

making channel driving our supply shock channel results.

³²Our findings remain very similar if we use $\pm 1\%$ as the extreme return in our definition and in this case, we end up 42 parent-order executions with *KnightSameSide*=1.

significant at the level of 1% but is relatively small when contrasted to the increase in institutional trading costs. The increase in effective spreads is mostly driven by the price impact component, even though the increase in price impact is not itself statistically significant. Large fluctuations in prices during the period of the glitch make, however, the interpretation of the standard five-minute spread decomposition in (5) and (6) subject to caution, an issue we further discuss in Section 6.2.

More importantly, effective spreads do not pick up the large increase in average institutional trading costs on glitch-affected stocks in the aftermath of the glitch. Price impact displays a small and statistically significant increase. This suggests that price impact may align better with institutional trading costs than effective spreads. We show in Section 6.2, however, that all the spread-based measures exhibit only little relation to institutional costs measures in our sample.

Effective spreads do not exhibit any increase on the day of the glitch for DMM stocks. One should keep in mind, as mentioned above, that the DMM tests may lack power due to the small number of executions on DMM stocks in our benchmark sample (restricted to NYSE-listed S&P 500 stocks). Our sample focuses on stocks for which we have parent-order execution data; that is, NYSE-listed S&P 500 stocks. One drawback is that only 17 of these stocks have KC as designated market maker (DMM) and are not affected by the glitch. Furthermore, all of these stocks are large and actively-traded, which may limit the importance of the DMM for their liquidity. We show in Section 7.1 that the results differ when we use daily spreads for an extended sample of DMM stocks. We find that spreads of DMM stocks increase by 5.45 basis points on the day following the glitch. This increase is statistically significant at 10% level. There is no statistical evidence of an increase in spreads beyond the day following the glitch.

5.3. Discussion

Our first key finding is that a large shock to the inventory of a market participant has a persistent impact on institutional trading costs of glitch-affected stocks. The fact that the effect persists even after KC has sold off its portfolio to GS and even after its DMM responsibilities have been reassigned to GETCO (see Section 2) is strong evidence for the supply shock channel discussed in Section 2.3. Since we control for the participation rate and volatility, the speed at which liquidity reverts to normal is an indication that market making capital is constrained and not immediately redeployed, so that it takes time for the inventory risk to be spread efficiently across investors. We provide further evidence in support of the supply shock channel by showing that trading costs increase in an asymmetric manner: trades

are particularly costly if they are in the same direction as KC (or GS) expected order-flow. Instead, trades that effectively supply liquidity to KC (or GS) seeking to off-load their risk incur lower trading costs.

Our second key finding is that stocks with KC as DMM experience an increase in institutional trading costs on the day of the glitch. This supports the results of [Comerton-Forde et al. \(2010\)](#), who show that a specialist's inventories and trading revenues are positively related to the value-weighted spread among stocks assigned to that specialist. Our setup provides a clean natural experiment in a more recent sample with a different market structure. Since DMMs have fewer obligations than NYSE specialists for the stocks they supervise, it is unclear how important they are for the liquidity of their underlying stocks.

Our third key finding is that traditional spread-based microstructure measures of liquidity diverge from the institutional trading cost measures. This result highlights the multi-faceted nature of liquidity: a trading glitch, such as a 'flash-crash,' may have persistent adverse effects on the liquidity faced by institutional investors and these effects may be difficult to detect using standard (spread-based) liquidity measures. In the next section, we focus on this result and investigate why traditional spread-based measures do not reflect the increase in institutional trading costs.

6. Discrepancy between Standard Liquidity Measures and Institutional Costs Measures

Why do traditional spread-based measures not reflect the increase in institutional trading costs in the aftermath of the glitch? First, we examine the behavior of spreads on the day of the glitch and formally estimate the relation between spreads and institutional cost measures. Second, we study the dynamics of other liquidity measures around the glitch to see whether they align better with institutional cost measures.

6.1. Spreads on the Day of the Glitch

Figure 4 plots effective spread, realized spread, and price impact for glitch-affected on August 1. The glitch has a large but short-lived impact on the spreads of affected stocks (left plot). Effective spreads stabilize after 10:00 AM and do not exhibit marked variation for the rest of the day. Realized spreads spike during the glitch but then fall back rapidly. We have followed the literature and used an interval of five minutes to compute realized spread and price impact. An interval of five minutes may not be appropriate here given the duration of the glitch, and the large fluctuations in both measures should not be taken as evidence

against their validity in other contexts. In the Internet Appendix, we show that the measures behave more “smoothly” over the interval of the glitch if we extend the interval to compute them.³³

The right plot of Figure 4 compares effective spreads on affected stocks to effective spreads on nonaffected stocks matched by market capitalization and price. After 10:10 AM it is hard to discern any clear difference between the two series. Liquidity, as measured by spreads, appears to be resilient.

[Insert Figure 4 here]

6.2. Explaining Institutional Cost Measures with Spreads

So far our results suggest that Implementation Shortfall and VWAP Slippage do not measure the same dimension of liquidity as spread-based measures. This evidence is based, however, on a specific event (the glitch). To document formally this discrepancy we directly regress the institutional cost measures on effective spread, realized spread, and price impact along with our set of control variables. We exclude parent-order executions that occur between August 1, 2012 and August 14, 2012 (inclusive) but the regression estimates are very similar if we include this period.

Table 6 reports the results. As expected from the results in Table 4, we find only little correlation between the two sets of measures. Effective spread is not significantly correlated with either of the institutional trading cost proxies. Price impact displays a positive association with Implementation Shortfall, but the correlation is small and not statistically significant.

[Insert Table 6 here]

There are several potential reasons for the surprisingly small correlation between the two sets of measures. First, this result may be specific to our sample, though we note that [Eaton et al. \(2019\)](#) find a similar result using Abel Noser data. Second and more importantly, effective spread does not measure depth, which is likely of first-order importance for the actual implementation shortfall on a large trade. Indeed, we document below that total depth at the best bid and ask tend to decrease for affected stocks in the aftermath of the glitch.

³³[Conrad and Wahal \(2019\)](#) explore how changing the estimation interval affects realized spreads. They find that aggregate profits to liquidity provision (computed from realized spreads) are negative at horizons beyond one second in 2012.

6.3. Turnover and Depth around the Glitch

We examine next two other measures of liquidity: turnover and depth.

Turnover is often used as a measure of liquidity. Turnover and spreads are generally negatively correlated in the cross-section, which is intuitive. In contrast, Lee et al. (1993) find that turnover is positively associated with spreads in the time-series. Theoretically, this result can potentially be rationalized if an increase in turnover reflects a higher degree of adverse selection.

Independently of the interpretation of turnover as a measure of liquidity, as discussed in section 2.2., the supply-shock channel predicts that, after the trading glitch, trading activity should increase on glitch-affected stock, whereas the market-making capital channel predicts that trading activity should decrease on the DMM stocks. Hence, we expect to observe variations in turnover around the glitch across glitch-affected and non-affected DMM stocks.

Further, as pointed out by Domowitz et al. (2005) and others, the quoted spread only represents the inside quote for a small trade size. The effective spread improves on that as it is not restricted to the inside quote. However, it is still only an ex-post measure for a specific (typically small) trade size and does not correspond to an ex-ante measure of liquidity of a potential (large) trade size. Thus, especially for the large institutional investors in our sample, it may not capture all relevant aspects of liquidity, such as market depth. Indeed, for a given effective spread, we would expect the cost of trading a large order to be smaller if there are more liquidity providers on each side of the book, that is if market depth is larger. And it is plausible that market depth can change in a way that is not perfectly correlated with the effective spread.

We thus test whether increase in institutional trading costs is associated with a decrease in the available depth, not captured by the change in spreads.

Since we can compute depth and turnover every day for every stock in our sample, we estimate a slightly modified version of (7). The key difference is that the adjusted regression is not at the parent-order execution level but at the day-stock level. Let $C_{i,t}$ be the liquidity measure for stock i on day t . We estimate the following regression:

$$(8) \quad C_{i,t} = \sum_{k=0}^5 \delta_k \text{Glitch}_{k,i,t} + \delta_M \text{GlitchAug9ToAug31}_{i,t} + \sum_{k=0}^5 \gamma_k \text{DMM}_{k,i,t} \\ + \gamma_M \text{DMMAug9ToAug31}_{i,t} + \alpha_i + \alpha_t + \sum_j \beta_j \text{Control}_{j,i,t} + \epsilon_{i,t},$$

where $\text{Glitch}_{k,i,t}$ equals one on the k^{th} trading day after the glitch ($k = 0, \dots, 5$) if stock i was affected by the glitch and $\text{GlitchPost}_{i,t}$ equals one for affected stocks on any date more than five trading days after the glitch. $\text{DMM}_{k,i,t}$ equals one on the k^{th} trading days after the

glitch, if stock i has KC as DMM *and* is not affected by the glitch.

We decompose the five trading days after the glitch into five separate indicator variables. We also report results with a specification exactly similar to that of (7) in which we keep the single indicator variable for the five trading days after the glitch.

In addition to stock and day fixed effects, we use the following stock-specific control variables at each date: log price, log turnover (obviously not when turnover is used as $C_{i,t}$), log market capitalization, and volatility (volatility is computed based on daily high and low prices as in Parkinson (1980)). Liquidity variables and control variables are winsorized at 0.05% and 99.95%.

The results are reported in Table 7.³⁴ The first two columns report the results for daily effective spreads as a point of comparison. The results are in line with that of Table 4: Effective spreads of glitch-affected stocks spike on the day of the glitch but then show no pattern in the following days.

[Insert Table 7 here]

Columns 3 and 4 in Table 7 show that turnover spikes by 17% on average on the day of the glitch for glitch-affected stocks. This result is expected since the glitch generated an enormous amount of abnormal trading activity. The more interesting result is that the turnover of glitch-affected stocks increases substantially in the four days following the glitch. The increase is not statistically significant on all days but, when it is, the economic magnitude is large: an average increase of around 8-13% in each of the four days. The abnormal trading activity is consistent with the idea of KC/Goldman trying to unload the positions in the aftermath of the glitch.³⁵ Turnover is also slightly lower for DMM stocks in the two days following the glitch but not significantly so. A lower turnover is expected if KC reduces its market making activity in the stocks for which it is a market maker.

Columns 5 and 6 in Table 7 show that total depth at the best bid and best ask decreases significantly on the day of the glitch. The economic magnitude of this decrease in depth is large. And total depth remains much lower in the days following the glitch, though the coefficients are not statistically significant on these days. A persistent decrease in depth following the shock to the risk-bearing capacity of KC could help explain the discrepancy

³⁴Standard errors are clustered by firm. Double clustering by day and firm using the method of Thompson (2011) gives similar results, but several of the standard errors are not well defined. The coefficients on realized spreads and price impact may not sum up exactly to the coefficient on effective spreads because the liquidity measures are winsorized.

³⁵Since we include measures of turnover as controls in all of our tests, the increase in trading activity does not explain the increase in institutional trading costs. In contrast, following on the results of Lee et al. (1993), one may think that controlling for turnover could actually explain why we do not observe a persistent increase in spreads. However, we obtain similar results when we do not include turnover as a control.

between spread-based measures and institutional trading cost measures. It also suggests one could improve on the traditional spread-based measures by taking into account information on the depth of the limit order book.³⁶

Total depth at the best bid and best ask on DMM stocks does not display any strong pattern in the two days following the glitch. Curiously, total depth for these stocks is significantly lower three to five trading days after the glitch. Three business days after the glitch corresponds to the date when GETCO took over KC's market making duties. One potential explanation for the decrease in depth is that the sudden takeover of market making duties for hundreds of stocks forced GETCO to restrict the quantity of liquidity supplied on these stocks for at least a few days. We would not necessarily expect a simultaneous increase in spreads because designated market makers are restricted to quote at the best ask and best bid for a given fraction of the month.

6.4. Discussion

Traditional spread-based measures do not appear to detect variations in institutional trading costs. This absence of relation holds both over the full sample and in the aftermath of the glitch. Our results highlight that other measures of liquidity such as turnover and depth should be monitored in conjunction to spreads. This recognizes that liquidity is multifaceted.

7. Robustness Checks

This section presents several robustness checks related to our main results.

7.1. Extended Sample of DMM Stocks

To further test the market making capital theory, we focus on a larger sample of stocks, which includes all common stocks (CRSP share code 10 or 11) listed on the NYSE or NYSE MKT (CRSP exchange code 1 or 2) from June 1 to September 28, 2012 (84 days). Stocks are required to have a price greater than \$2 and lower than \$1000 at the end of June, a market capitalization greater than \$100 million at the end of June, and valid data for all variables every day of the sample. In total, we obtain data for more than 1,300 stocks. In this sample, 114 stocks are not affected by the trading glitch and have KC as DMM.

³⁶For example, one could compute a hypothetical IS-curve by running through the order book for different size orders. Unfortunately, we cannot compute such a IS-curve with our data, since in TAQ we only have the information on the available depth at the best bid or ask and not the entire order book.

We estimate a regression similar to (8) using this extended sample. Table 8 presents the results. The key coefficients of interest here are the DMM indicators, which take the value one around the glitch for stocks that have KC as DMM and are not affected by the glitch. On the day of the glitch DMM stocks tend to have higher spreads, but this increase is not statistically significant. However, DMM stocks tend to have substantially higher effective spreads—by 5.45 basis points—on the day following the glitch, which is statistically significant. The average increase in effective spreads remains positive for two more days but is statistically insignificant. The last column of the table confirms our previous results on turnover. The turnover of DMM stocks is lower in the two days following the glitch and significantly so in the second day.

This evidence supports the market making capital theory. The negative shock to KC's risk-capital adversely affected its ability to fulfill its role as designated market maker in a number of assigned stocks. The results also suggest that the DMM is less important for S&P 500 stocks than for smaller stocks since we do not observe a similar increase in effective spread for DMM stocks in Table 7. This result is intuitive but remains a conjecture without additional data on DMM activity.

[Insert Table 8 here]

7.2. Addressing Alternative Explanations

One alternative explanation for our findings could be related to potential failure of the Bank's execution during the glitch period. If the Bank did a worse job for its customers during the glitch day and the subsequent week, one would expect that most of these customers would not use this broker for their future trading given the competition in this area. There are 110 unique investors who sent at least one order to this broker between Aug 1, 2012 and Aug 8, 2012, inclusive. We find that 105 of these investors sent an order to the broker in the post-glitch period suggesting that investors were not in any way discouraged by the performance of the broker.

Another concern would be the potential change in the customer flow. For example, if investors who are affected by the glitch, start trading with the Bank, the persistent cost increase could be explained by their urgent trading needs. This concern is invalidated by the construction of our data set as explained in Appendix B. An investor trading during the glitch period must be trading with the bank prior to June 2012 as otherwise their parent-order executions would not be observed in the data set. Thus, there are no new customers appearing during the glitch period in our data set which rules out this alternative hypothesis.

7.3. Shorter Sample Centered around the Glitch

We use the full year 2012 in our main regressions to increase the power of our tests and better contrast institutional liquidity measures to standard liquidity measures. As reported in Table 9, our main results are robust to using a 4-month window around the date of the trading glitch.

[Insert Table 9 here]

7.4. Extended List of Affected Stocks

As a robustness check, we consider two alternative methods to classify glitch-affected stocks. First, we quantify how abnormal is the number of trades compared to its recent average. Using annual data between August 1, 2011 and July 31, 2012, we first compute $RATIO_{i,t} \triangleq \frac{NumTr_{i,t}}{RollNumTr_{i,t}}$ for each stock and trading day where $NumTr$ denotes the number of trades between 9:30 AM and 10:00 AM, and $RollNumTr$ denotes the mean of this statistic over the prior five trading days. At the stock level, we compute the mean and standard deviation of $RATIO$. We then compute the standardized z-score of $RATIO_{i,t}$ using these historical moments at the stock level. We define a stock as glitch-affected, if it has a standardized z-score of 5 or more. We obtain 47 stocks as glitch-affected with this definition. Compared to our original list in Table 3, COH becomes unaffected and 11 stocks get added.

[Insert Table 10 here]

Table 10 illustrates that our regression results are very similar to those of the main analysis in Section 5.1. Both cost metrics suggest that affected parent-order executions exhibit a highly statistically significant cost increase on the day of the trading glitch. The estimated cost increase due to the trading glitch is approximately 5 (20) bps for VWAP Slippage (IS). For both cost measures, parent-order executions occurring in the following week (within the next 5 business days) are exposed to a cost increase of a roughly similar magnitude. As measured by VWAP Slippage (IS), parent-order executions in the affected group realize on average an additional 5 bps (16 bps) in the following week which are similar to the estimates obtained earlier.

Second, we utilize the list of potentially affected stocks provided directly by NYSE. At 11:48am on August 1, 2012, NYSE released a trader update stating that NYSE and NYSE MKT are reviewing trades in several symbols, 49 of which are S&P 500 stocks. In Section 2 of the Internet Appendix, we show that some of these stocks do not appear to be subject to any abnormal trading activity on August 1. Nevertheless, we rerun our analysis using this

list of stocks as glitch-affected and our results remain largely similar as shown in Table IA.4 of the Internet Appendix.

7.5. Inclusion of NASDAQ-Listed Stocks

We did not include NASDAQ stocks in our analysis to avoid issues related to differences in market structure. In particular, NASDAQ stocks do not have designated market makers, which makes a comparison of their liquidity costs to that of NYSE-listed stocks difficult for the purpose of this paper. As a robustness check, Table 11 shows that our main results are unaffected by the inclusion of parent-order executions on NASDAQ-listed S&P 500 stocks in the sample.

[Insert Table 11 here]

7.6. Alternative Specifications and Consistency with Abel Noser Data

In unreported analyses, we have checked the robustness of our findings to alternative specifications. First, our results relative to DMM stocks are unaffected by excluding all glitch-affected stocks from the sample. Second, our results for glitch-affected stocks are robust to excluding all DMM stocks from the sample. Finally, our findings are also unaffected if in the control group, we use parent-orders on stocks matched by propensity scores based on market capitalization and turnover realized on July 31, 2012.

Finally, we obtain very similar findings if we run the regression in Equation (7) using large institutional parent-order executions from Abel Noser data around the glitch period. We include parent-order executions on NYSE-listed stocks occurring between July 24th, 2012 and August 8th, 2012 and exclude small parent-orders having a value lower than \$100 thousand dollars. There are approximately 95,000 parent-order executions that satisfy this criteria. This sample is very similar to our institutional trading data set with regards to the coverage of glitch- and DMM-affected stocks. There are 35 unique stocks affected by the glitch and 19 distinct stocks have KC as their DMM. Our set of control variables includes fraction of daily volume, logarithm of market capitalization, daily turnover and broker dummies. Similar to earlier findings, we find that for glitch-affected stocks institutional trading costs are economically and statistically significantly higher on the day of the trading glitch and in the subsequent week. The estimated cost increase in IS is approximately 3 bps on the glitch day and 5 bps for the subsequent five business days and both estimates are significant at 1% level. Consistent with our earlier results, we do not find any significant change in the execution costs of DMM stocks in the post-glitch period. Overall, these findings highlight

the robustness of our findings with regards to the institutional trading data set and reinforces the generalizability of our conclusions.

8. Conclusion

We examine the impact of an exogenous trading glitch at a high-frequency market-making firm on institutional trading costs (Implementation Shortfall and VWAP Slippage) obtained from a proprietary data set and on standard measures of stock liquidity (spreads, depth, and turnover).

We find that stocks that are heavily bought (sold) increase (decrease) by about 4% during the glitch and become substantially more illiquid. Prices take a full day to revert to their pre-opening level. The symmetric ‘wave-like’ price pattern we observe for bought versus sold stocks, which overshoot their pre-opening level an hour before market close, is reminiscent of the predictions of the slow-moving capital model presented in [Duffie \(2010\)](#). It is also qualitatively consistent with the price-pressure evidence around mutual fund fire-sales ([Coval and Stafford, 2007](#); [Edmans et al., 2012](#)). However, quantitatively our results suggest much ‘faster moving’ capital than these papers on fire sales. Indeed, we find that the 4% price pressure effect reverts within a day, whereas the aforementioned papers find that it takes several months.

When it comes to the liquidity of the glitch-affected stocks, we find that effective spreads revert quickly, suggesting that market liquidity is resilient. In contrast, institutional trading costs remain significantly higher for more than one week. This cost increase is economically significant when compared to the cost of a typical large-order trade. These results also hold in a completely different institutional trading data set aggregated by Abel Noser.

Trading costs are larger for orders executed in the same direction as the market-making firm who wants to unload its abnormal inventory after the glitch. This asymmetric impact on market liquidity is consistent with theories where, because of anticipated supply shocks or predatory trading, current inventory shocks have persistent effects on trading costs.

Our results indicate a substantial discrepancy between institutional trading cost measures and standard microstructure spread-based liquidity measures. We investigate why the persistent cost increase in institutional trading costs is not captured by effective spreads. We find that stocks affected by the glitch experience abnormal turnover and lower market depth in the week following the glitch. These findings suggest that standard liquidity measures may not pick up the dimension of liquidity that is affected by the glitch. Importantly, we document that the discrepancy between institutional trading cost measures and standard spread-based liquidity measures is not restricted to the glitch period but extends to our full

sample. This suggests that these two types of measures capture different facets of market liquidity, some of which may be more relevant for institutional investors. Further research is warranted to better understand the discrepancy between the two types of liquidity measures.

Finally, we document that all stocks for which the firm was a designated market maker become more illiquid on the day of the glitch, even though these stocks were not heavily traded during the glitch. Furthermore, we also provide evidence that effective spreads on these stocks that are not in the S&P 500 increase significantly in the aftermath of the glitch. Since KC suffers a capital shock, its incentive to provide liquidity drops as predicted by theory. The decline in liquidity is more persistent for stocks that are less actively traded.

Overall, our empirical evidence is broadly consistent with ‘slow-moving capital’ theories, however our finding that ‘supply’ shocks have a more persistent effect on a stock’s illiquidity than on its price level, seems hard to reconcile with predictions of standard models and thus warrants further research.

References

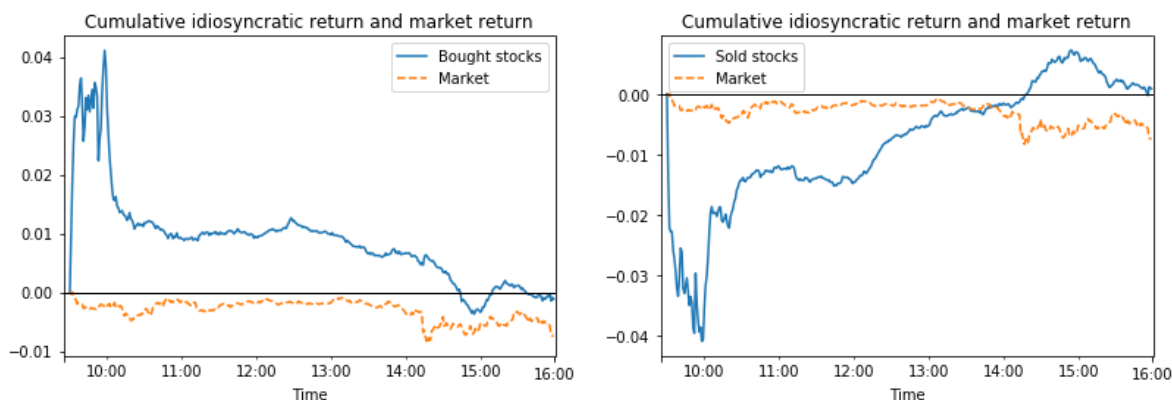
- Amihud, Y. and H. Mendelson (1980). Dealership market. *Journal of Financial Economics* 8(1), 31–53. [1](#)
- Anand, A., P. Irvine, A. Puckett, and K. Venkataraman (2011). Performance of institutional trading desks: An analysis of persistence in trading costs. *Review of Financial Studies* 25(2), 557–598. [17](#)
- Barra (1997). Market impact model handbook. *BARRA Inc., Berkeley*. [3](#), [11](#)
- Bessembinder, H., J. Hao, and K. Zheng (2019). Liquidity Provision Contracts and Market Quality: Evidence from the New York Stock Exchange. *Review of Financial Studies*. [4](#)
- Bogousslavsky, V. (2016). Infrequent rebalancing, return autocorrelation, and seasonality. *Journal of Finance* 71(6), 2967–3006. [10](#), [56](#), [57](#)
- Brogaard, J., T. Hendershott, and R. Riordan (2014). High-frequency trading and price discovery. *Review of Financial Studies* 27(8), 2267–2306. [6](#), [24](#)
- Brunnermeier, M. K. and L. H. Pedersen (2005). Predatory trading. *Journal of Finance* 60(4), 1825–1863. [11](#)
- Clark-Joseph, A. D., M. Ye, and C. Zi (2017). Designated market makers still matter: Evidence from two natural experiments. *Journal of Financial Economics* 126(3), 652–667. [4](#)
- Comerton-Forde, C., T. Hendershott, C. M. Jones, P. C. Moulton, and M. S. Seasholes (2010). Time variation in liquidity: The role of market-maker inventories and revenues. *Journal of Finance* 65(1), 295–331. [1](#), [4](#), [5](#), [12](#), [28](#)
- Conrad, J. S. and S. Wahal (2019). The term structure of liquidity provision. *Journal of Financial Economics*, forthcoming. [29](#)
- Coughenour, J. F. and M. M. Saad (2004). Common market makers and commonality in liquidity. *Journal of Financial Economics* 73(1), 37–69. [12](#)
- Coval, J. and E. Stafford (2007). Asset fire sales (and purchases) in equity markets. *Journal of Financial Economics* 86(2), 479–512. [2](#), [10](#), [11](#), [36](#)
- Demsetz, H. (1968). The cost of transacting. *Quarterly Journal of Economics*, 33–53. [1](#)

- Domowitz, I., J. Glen, and A. Madhavan (2001). Liquidity, volatility and equity trading costs across countries and over time. *International Finance* 4(2), 221–255. [21](#)
- Domowitz, I., O. Hansch, and X. Wang (2005). Liquidity commonality and return co-movement. *Journal of Financial Markets* 8(4), 351–376. [1](#), [30](#)
- Duffie, D. (2010). Presidential address: Asset price dynamics with slow-moving capital. *Journal of Finance* 65(4), 1237–1267. [1](#), [2](#), [5](#), [9](#), [10](#), [11](#), [36](#), [56](#)
- Eaton, G. W., P. J. Irvine, and T. Liu (2019). Measuring institutional trading costs and the implications for finance research: The case of tick size reductions. *Available at SSRN* 3255735. [5](#), [29](#)
- Edmans, A., I. Goldstein, and W. Jiang (2012). The real effects of financial markets: The impact of prices on takeovers. *Journal of Finance* 67(3), 933–971. [2](#), [10](#), [11](#), [36](#)
- Grossman, S. J. and M. H. Miller (1988). Liquidity and Market Structure. *Journal of Finance* 43(3), 617–633. [1](#), [9](#), [12](#)
- Hendershott, T., C. M. Jones, and A. J. Menkveld (2011). Does algorithmic trading improve liquidity? *Journal of Finance* 66(1), 1–33. [6](#)
- Hendershott, T. and A. J. Menkveld (2014). Price pressures. *Journal of Financial Economics* 114(3), 405–423. [1](#)
- Hendershott, T. and M. S. Seasholes (2007). Market maker inventories and stock prices. *American Economic Review* 97(2), 210–214. [1](#)
- Holden, C. W. and S. Jacobsen (2014). Liquidity measurement problems in fast, competitive markets: Expensive and cheap solutions. *Journal of Finance* 69(4), 1747–1785. [17](#)
- Hu, G., K. Jo, Y. A. Wang, and J. Xie (2017). Institutional trading and abel noser data. *Available at SSRN*. [18](#)
- Johnson, N., G. Zhao, E. Hunsader, H. Qi, N. Johnson, J. Meng, and B. Tivnan (2013). Abrupt rise of new machine ecology beyond human response time. *Scientific Reports* 3, 2627. [6](#)
- Jones, C. M. and M. L. Lipson (2001). Sixteenths: direct evidence on institutional execution costs. *Journal of Financial Economics* 59(2), 253–278. [5](#)
- Kirilenko, A. A., A. S. Kyle, M. Samadi, and T. Tuzun (2017). The flash crash: High-frequency trading in an electronic market. *Journal of Finance* 72(3), 967–998. [6](#)

- Korajczyk, R. A. and D. Murphy (2018). High-frequency market making to large institutional trades. *Review of Financial Studies* 32(3), 1034–1067. [18](#)
- Lee, C. and M. J. Ready (1991). Inferring trade direction from intraday data. *Journal of Finance* 46(2), 733–746. [17](#)
- Lee, C. M. C., B. Mucklow, and M. J. Ready (1993). Spread, Depths, and the Impact of Earnings Information: An Intraday Analysis. *Review of Financial Studies* 6(2), 345–374. [30](#), [31](#)
- Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business* 53, 61–65. [31](#), [46](#), [51](#), [52](#)
- Perold, A. F. (1988). The implementation shortfall: Paper versus reality. *Journal of Portfolio Management* 14(3), 4–9. [1](#), [13](#)
- Petersen, M. A. (2009). Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies* 22(1), 435–480. [22](#)
- Puckett, A. and X. S. Yan (2011). The interim trading skills of institutional investors. *The Journal of Finance* 66(2), 601–633. [17](#)
- Sağlam, M., C. Moallemi, and M. Sotiropoulos (2018). Short-term trading skill: An analysis of investor heterogeneity and execution quality. *Journal of Financial Markets*. [57](#)
- Thompson, S. B. (2011). Simple formulas for standard errors that cluster by both firm and time. *Journal of Financial Economics* 99(1), 1–10. [31](#)
- van Kervel, V. and A. J. Menkveld (2017). High-frequency trading around large institutional orders. *Journal of Finance*. [18](#)
- Wardlaw, M. (2019). Measuring mutual fund flow pressure as shock to stock returns. working paper. [3](#), [11](#)

Figure 1: Cumulative return and share volume of bought and sold stocks on August 1, 2012. This figure plots the average cumulative idiosyncratic return and total share volume of glitch-affected stocks that are bought (9:30-10:00 AM return $> 1\%$) and sold (9:30-10:00 AM return $< -1\%$) over the interval of the glitch. For each stock, idiosyncratic returns are the residuals of a regression of the stock's one-minute log return on the value-weighted log market return of all stocks in the sample excluding glitch-affected stocks on August 1, 2012 (without an intercept). Panel (a) reports the cumulative idiosyncratic return and market return. Share volume is summed every minute across all bought/sold stocks and divided by the total share volume of all stocks in the sample excluding glitch-affected stocks. The summed volumes are normalized by their averages over the previous month. Panel (b) reports the 15-minute rolling average normalized share volume.

(a) Cumulative return on August 1, 2012



(b) Share volume of bought/sold stocks relative to other stocks on August 1, 2012

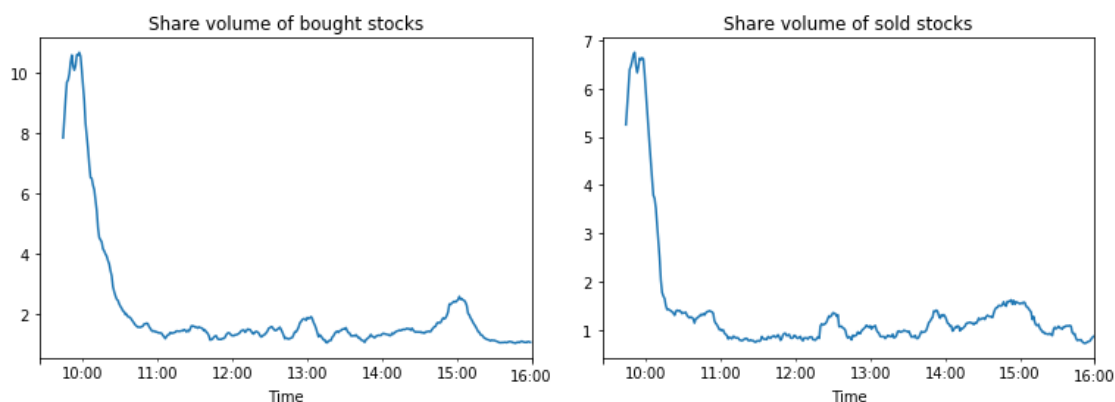


Figure 2: Price paths predicted by various models with a positive or a negative supply shock. Details and calibrations of the inventory and slow-moving capital models are provided in the appendix.

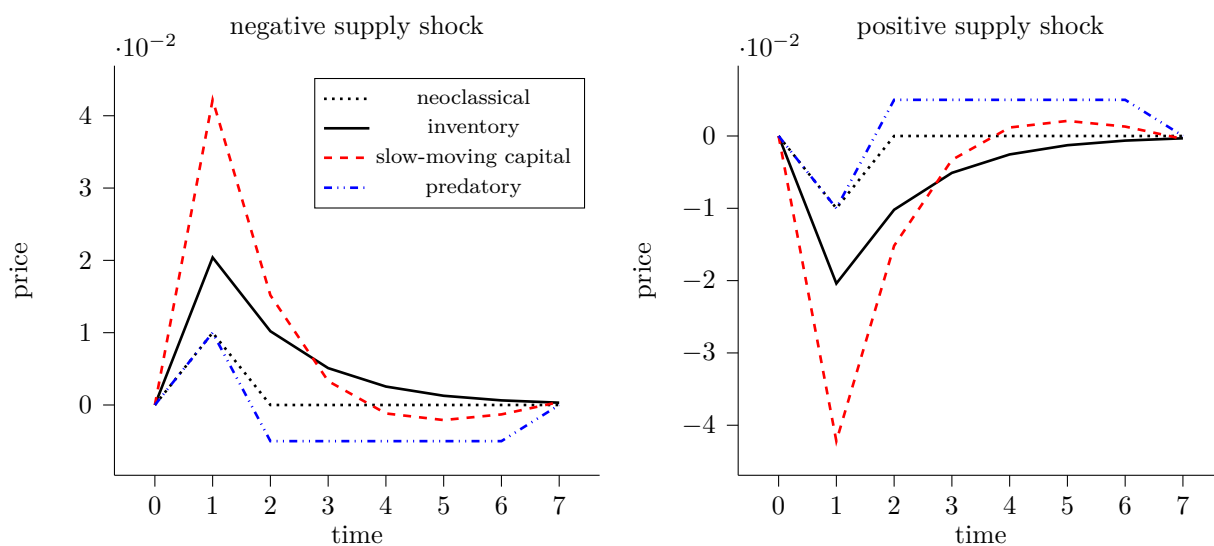


Figure 3: Difference in VWAP slippage between the affected and unaffected groups of stocks during the period of July 2012 and September 2012 (inclusive). We report the mean and the median values across different periods including the glitch period.

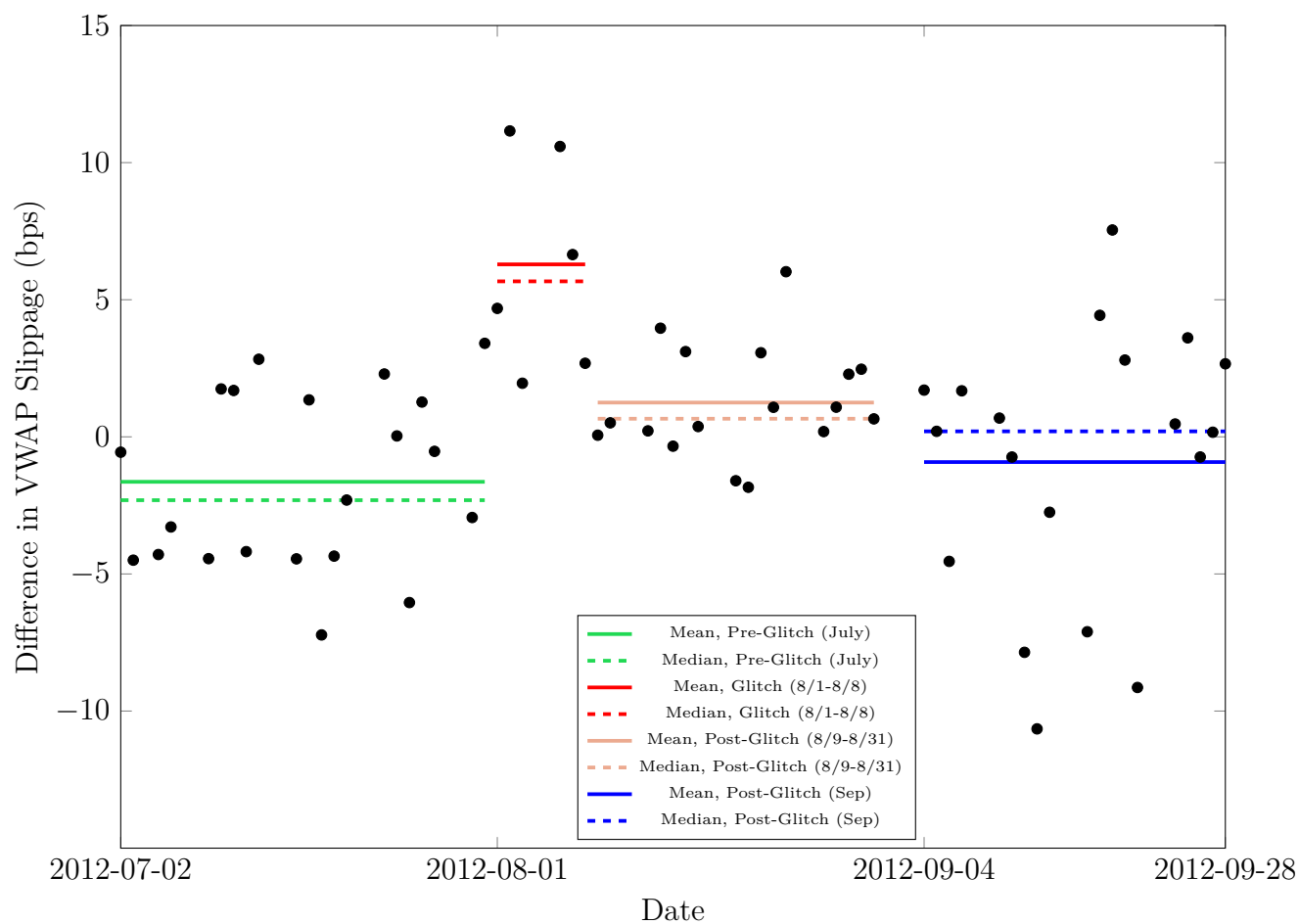


Figure 4: Intraday effective spread, realized spread, and price impact on August 1, 2012. For each stock, liquidity measures for all transactions are averaged using dollar-weighting over non-overlapping 1-minute intervals starting at 9:30 AM. The plotted liquidity measure in a given interval is computed as the cross-sectional median in the interval. The left plot shows the liquidity measures for stocks affected by the glitch. The right plot compares effective spreads on affected stocks to effective spreads on control stocks that are matched by market capitalization and price at the beginning of June.

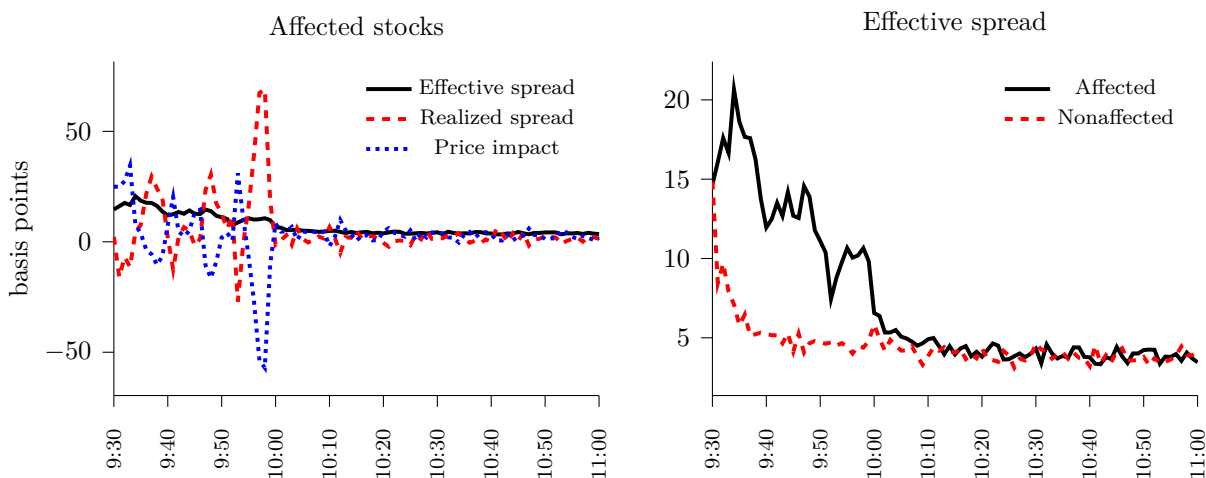


Table 1: Summary statistics for institutional executions. The data set is composed of institutional executions on NYSE-listed S&P 500 stocks from January 2012 to December 2012. The participation rate is the ratio of order size to the total volume during the lifetime of the parent-order execution. Average daily volatilities are computed using the previous 20 trading days before the parent-order execution date. Panel (a): All parent-order executions (31,792). Panel (b): Parent-order executions implemented on August 1, 2012 (534).

(a) All parent-order executions between January 2012 and December 2012

	Mean	St. Dev.	Min	Q1	Med.	Q3	Max
Order Size (\$ M)	0.92	2.00	<0.01	0.08	0.34	1.02	66.50
Participation Rate (%)	5.34	7.88	<0.01	0.13	0.95	9.16	77.92
Implementation Shortfall (%)	0.03	0.61	-10.06	-0.18	0.03	0.23	9.97
VWAP Slippage (%)	0.00	0.12	-3.00	-0.01	0.01	0.03	2.35
Average Daily Volatility (%)	1.19	0.43	0.13	0.88	1.11	1.41	4.20
Fraction of Trading Time (%)	40.75	40.41	1.28	4.73	17.77	92.53	100.00
Interval Return (%)	0.01	0.81	-11.63	-0.29	0.00	0.30	11.09

(b) Parent-order executions on August 1, 2012

	Mean	St. Dev.	Min	Q1	Med.	Q3	Max
Order Size (\$ M)	0.16	0.48	<0.01	0.01	0.03	0.06	4.74
Participation Rate (%)	1.34	3.18	0.01	0.18	0.52	1.07	29.21
Implementation Shortfall (%)	0.001	0.32	-1.81	-0.12	-0.003	0.08	4.07
VWAP Slippage (%)	-0.005	0.12	-1.39	-0.03	0.004	0.03	1.39
Average Daily Volatility (%)	1.30	0.47	0.61	0.95	1.18	1.57	3.33
Fraction of Trading Time (%)	11.11	20.51	1.33	3.31	3.72	15.34	100.00
Interval Return (%)	-0.11	0.42	-2.15	-0.25	-0.08	0.06	4.46

Table 2: Summary statistics for daily liquidity measures and other variables. Daily spreads are computed by dollar-weighting intraday transaction spreads. Total depth is the sum of the time-weighted depth at the best bid and the time-weighted depth at the best ask. Price is the daily closing price. Volatility is computed based on daily high and low prices as in [Parkinson \(1980\)](#). The sample consists of NYSE-listed S&P 500 stocks from January 2012 to December 2012, excluding days with an early closure (07/03, 11/23, and 12/24). The total number of observations is 96,552 allocated over 394 stocks. The descriptive statistics are calculated across all stock-day observations. Spreads are winsorized at 0.05% and 99.95%. The within standard deviation is computed using deviations from the time mean for each stock.

	Mean	Q1	Med.	Q3	St. Dev. (within)
Effective Spread (bp)	4.41	2.60	3.42	4.86	2.36
Realized Spread (bp)	2.38	0.79	1.80	3.19	3.55
Price Impact (bp)	1.89	0.52	1.55	2.86	2.98
Total Depth (\$1,000)	306.82	84.71	135.14	275.53	282.62
Turnover (%)	1.03	0.51	0.75	1.18	0.78
Dollar Volume (\$million)	175.69	57.51	104.33	197.17	135.62
Price	53.91	27.95	44.01	65.49	6.35
Market Cap. (\$billion)	24.95	6.69	12.42	24.48	3.05
Volatility (%)	1.22	0.74	1.04	1.48	0.60

Table 3: Composition of the sample. Panel (a) reports stocks that we identify as being affected by the glitch. The panel shows the number of trades for the different stocks between 9:30 and 10:00 AM on August 1, 2012 (AUG1), as well as the average number of trades in the same interval over the previous five trading days (AVG). RATIO is the ratio of AUG1 over AVG. Z denotes the Z-score of RATIO, which is measured in terms of standard deviations from the mean. The mean and standard deviation of RATIO are computed using data from August 1, 2011 to July 31, 2012. A stock is labeled as glitch-affected if it has more trades than the most active S&P 500 ETF (SPY) or if its RATIO is larger than 10. Panel (b) allocates the number of stocks in our sample among those that are identified as being affected by the glitch and those for which Knight Capital is the designated market maker (DMM). Sample: NYSE-listed S&P 500 stocks on August 1, 2012.

(a) List of stocks affected by the glitch

Ticker	AUG1	AVG	RATIO	Z	Ticker	AUG1	AVG	RATIO	Z
GNW	57726	1281.0	45.1	60.0	SWN	39628	2769.2	14.3	14.9
FRX	26703	637.6	41.9	58.0	LH	6081	429.0	14.2	16.1
HOG	40318	1044.0	38.6	44.6	AMD	34325	2996.0	11.5	11.0
AMT	56804	1533.4	37.0	65.1	LSI	81165	7298.8	11.1	12.2
EXC	58305	1576.6	37.0	44.0	WLP	55149	5054.2	10.9	12.3
GME	31571	925.0	34.1	24.9	F	109102	10432.0	10.5	15.5
JWN	37570	1302.8	28.8	30.7	PEP	38706	3856.0	10.0	14.6
MPC	36460	1558.3	23.4	25.6	HUM	31778	3261.6	9.7	9.7
AA	102620	4434.6	23.1	44.4	GT	41137	4395.0	9.4	10.9
AGN	36441	1704.2	21.4	28.1	CVS	34523	3961.8	8.7	9.5
LOW	94178	4911.2	19.2	34.8	BAC	124196	15921.6	7.8	13.1
BRKB	36994	2033.6	18.2	15.7	VZ	33336	5457.4	6.1	10.8
JNPR	169111	9381.4	18.0	18.6	C	91238	14958.4	6.1	10.4
AFL	33834	2043.0	16.6	29.2	WFC	71688	11892.8	6.0	12.4
ANF	35805	2305.2	15.5	10.8	DOW	32304	5843.0	5.5	8.8
DD	39901	2586.0	15.4	20.0	ANR	42395	7669.4	5.5	8.0
GLW	80269	5285.0	15.2	20.8	PFE	67551	15104.0	4.5	8.8
BBY	60240	3997.2	15.1	13.3	COH	39728	9374.0	4.2	3.0
DE	43236	2972.8	14.5	23.2					

(b) Composition of the sample between glitch-affected stocks and DMM stocks

Number of stocks		
Glitch	DMM	1
	no DMM	36
no Glitch	DMM	17
	no DMM	340
Total		394

Table 4: Panel regressions of institutional liquidity measures and matched standard liquidity measures. This table reports the results of regressing cost measures on indicator variables, control variables, and day and stock fixed effects using institutional parent-order executions on NYSE-listed S&P 500 stocks from January 2012 to December 2012. For each institutional parent-order execution, standard liquidity measures and control variables are computed over the interval of the execution as described in Section 4.4. *GlitchDay0* equals one if the parent-order execution is made on a glitch-affected stock on the day of the glitch. *GlitchNextWeek*, *GlitchAug9ToAug31*, and *GlitchPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. *DMM0* equals one if the parent-order execution is made on a stock for which Knight Capital is the designated market maker on the day of the glitch. *DMMNextWeek*, *DMMAug9ToAug31*, and *DMMPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. The liquidity measures are defined in Section 3. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

<i>Dependent variable:</i>	VWAP Slippage	Implementation Shortfall	Effective Spread	Realized Spread	Price Impact
GlitchDay0	6.35*** (1.77)	18.15** (8.17)	0.86*** (0.27)	-0.44 (0.56)	1.31* (0.71)
GlitchNextWeek	6.48*** (1.69)	16.66** (7.24)	0.41 (0.31)	-1.06 (0.74)	1.47* (0.81)
GlitchAug9ToAug31	1.11** (0.53)	-1.90 (3.27)	0.29 (0.25)	0.06 (0.20)	0.22 (0.32)
GlitchPost	0.37 (0.56)	-0.89 (2.45)	0.20 (0.27)	-0.02 (0.26)	0.22 (0.32)
DMM0	5.04*** (1.00)	32.92*** (8.51)	-0.09 (0.12)	-0.99*** (0.32)	0.90*** (0.32)
DMMNextWeek	-3.09 (2.88)	5.43 (5.73)	-0.005 (0.23)	-0.62 (0.67)	0.61 (0.60)
DMMAug9ToAug31	0.03 (1.07)	-3.47 (2.87)	-0.05 (0.19)	0.07 (0.30)	-0.11 (0.41)
DMMPost	0.02 (0.81)	-3.52 (5.36)	-0.04 (0.16)	0.08 (0.26)	-0.12 (0.31)
Participation Rate	14.76*** (2.17)	35.57*** (6.68)	0.88** (0.42)	-0.85 (1.01)	1.73* (1.02)
Fraction of Daily Volume	25.79*** (6.08)	128.84*** (49.04)	-6.90*** (1.20)	1.82 (2.39)	-8.72*** (2.23)
Relative Daily Volume	-0.27 (0.50)	-0.33 (1.72)	-0.26*** (0.07)	-0.24*** (0.08)	-0.02 (0.10)
IsEarningsDay	0.07 (0.90)	8.76** (4.17)	-0.11 (0.13)	-0.19 (0.31)	0.08 (0.34)
Interval Turnover	0.26*** (0.08)	0.30 (0.40)	-0.01 (0.01)	0.02 (0.01)	-0.03* (0.02)
Interval Volatility	9.00 (30.21)	63.73 (130.34)	200.99*** (45.05)	31.34 (27.75)	169.65*** (55.47)
Day Volatility	-80.15 (59.83)	-821.77*** (312.02)	-17.50 (11.90)	25.88 (20.46)	-43.39* (25.39)
Execution Duration Fraction	2.99*** (0.76)	5.42 (3.59)	0.43*** (0.12)	0.19 (0.17)	0.24 (0.17)
Day Return	20.79* (11.10)	-161.92** (72.19)	-2.13** (1.08)	-6.25** (2.78)	4.12 (2.89)
abs(Day Return)	-7.24 (19.79)	276.56** (118.66)	-2.51 (2.49)	-16.06*** (4.02)	13.56*** (4.77)
IsVWAP	-3.27*** (0.37)	-7.47*** (1.21)	0.004 (0.07)	-0.27 (0.16)	0.27 (0.18)
Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	31,792	31,792	31,641	31,641	31,641
Adjusted R ²	0.10	0.04	0.61	0.02	0.15

Table 5: Directional trading. This table reports the results of regressing institutional cost measures on indicator variables, control variables, and day and stock fixed effects using institutional parent-order executions on NYSE-listed S&P 500 stocks from January 2012 to December 2012. For each institutional parent-order execution, control variables are computed over the interval of the parent-order execution as described in Section 4.4. *KnightSameSide* is a dummy variable included to account for any cost increase in the stock executions for which KC is expected to trade in the same direction. Other variables are defined in Table 4. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

	VWAP Slippage	Implementation Shortfall
KnightSameSide	8.68** (4.25)	30.52* (18.47)
GlitchDay0	4.38** (1.90)	11.24 (8.23)
GlitchNextWeek	5.34*** (1.77)	12.64** (6.20)
GlitchAug9ToAug31	1.12** (0.54)	-1.88 (3.28)
GlitchPost	0.36 (0.56)	-0.91 (2.45)
DMM0	4.96*** (1.00)	32.62*** (8.48)
DMMNextWeek	-3.11 (2.89)	5.37 (5.68)
DMMAug9ToAug31	0.03 (1.07)	-3.46 (2.87)
DMMPost	0.03 (0.81)	-3.50 (5.36)
Participation Rate	14.77*** (2.17)	35.59*** (6.68)
Fraction of Daily Volume	25.82*** (6.09)	128.93*** (49.02)
Relative Daily Volume	-0.26 (0.50)	-0.30 (1.72)
IsEarningsDay	0.09 (0.90)	8.85** (4.17)
Interval Turnover	0.26*** (0.08)	0.30 (0.40)
Interval Volatility	10.13 (29.91)	67.69 (130.16)
Day Volatility	-88.10 (57.88)	-849.71*** (309.52)
Execution Duration Fraction	3.00*** (0.76)	5.45 (3.59)
Day Return	20.83* (11.13)	-161.78** (72.24)
abs(Day Return)	-6.66 (19.58)	278.61** (118.58)
IsVWAP	-3.27*** (0.37)	-7.48*** (1.21)
Fixed Effects	Yes	Yes
Observations	31,792	31,792
Adjusted R ²	0.10	0.04

Table 6: Explaining institutional cost measures with spreads outside of the glitch period. This table reports the results of regressing VWAP Slippage and Implementation Shortfall on effective spread, price impact, realized spread, control variables, and day and stock fixed effects using institutional parent-order executions on NYSE-listed S&P 500 stocks from January 2012 to December 2012. We exclude parent-order executions that occur between August 1, 2012 and August 14, 2012 (inclusive). For each institutional execution, spreads and control variables are computed over the interval of the parent-order execution as described in Section 4.4. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

	<i>Dependent variable:</i>					
	VWAP Slippage			Implementation Shortfall		
	(1)	(2)	(3)	(4)	(5)	(6)
Effective Spread	0.05 (0.05)			−0.71 (0.44)		
Price Impact		−0.08*** (0.02)			0.07 (0.12)	
Realized Spread			0.10*** (0.02)			−0.17** (0.08)
Participation Rate	14.63*** (1.07)	14.84*** (1.09)	14.77*** (1.08)	36.90*** (4.91)	36.05*** (4.93)	36.03*** (4.92)
Fraction of Daily Volume	29.05*** (4.72)	27.88*** (4.74)	28.48*** (4.74)	128.27*** (42.56)	134.18*** (42.69)	133.86*** (42.67)
Relative Daily Volume	−0.15 (0.50)	−0.16 (0.50)	−0.14 (0.50)	0.34 (1.53)	0.52 (1.53)	0.48 (1.53)
Interval Turnover	0.23*** (0.08)	0.22*** (0.08)	0.22*** (0.08)	0.21 (0.42)	0.22 (0.42)	0.22 (0.42)
Interval Volatility	28.11 (19.86)	55.86*** (16.64)	38.35** (17.13)	231.93** (102.53)	71.23 (114.20)	86.08 (124.45)
Day Volatility	−84.36 (63.42)	−91.46 (63.56)	−90.19 (63.41)	−946.87*** (315.45)	−926.05*** (316.29)	−922.55*** (316.49)
Execution Duration Fraction	3.22*** (0.58)	3.27*** (0.58)	3.23*** (0.58)	6.07* (3.17)	5.73* (3.18)	5.78* (3.18)
Day Return	17.59* (9.57)	17.86* (9.58)	18.11* (9.58)	−199.54*** (47.81)	−198.43*** (47.86)	−199.25*** (47.92)
abs(Day Return)	1.34 (19.05)	2.41 (19.12)	2.78 (19.07)	285.16*** (80.08)	285.61*** (80.22)	283.80*** (80.14)
IsVWAP	−3.20*** (0.21)	−3.17*** (0.21)	−3.17*** (0.21)	−7.07*** (0.90)	−7.10*** (0.90)	−7.13*** (0.90)
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	30,294	30,294	30,294	30,294	30,294	30,294
Adjusted R ²	0.10	0.10	0.10	0.05	0.04	0.04

Table 7: Standard liquidity measures around the glitch. This table reports the results of regressing daily percentage effective spread (ES), log turnover (log turn), and total dollar depth on indicator variables, day and stock fixed effects, and several control variables. *GlitchDay0* equals one for glitch-affected stocks on the day of the glitch. *GlitchNextWeek* and *GlitchAug9ToAug31* are defined in a similar way for glitch-affected stocks within the next 5 business days after the glitch and between August 9th and August 31st (included), respectively. *DMM0* equals one for stocks whose designated market maker is Knight Capital on the day of the glitch. *DMMNextWeek* and *DMMAug9ToAug31* are defined in a similar way for stocks whose designated market maker is Knight Capital within the next 5 business days after the glitch and between August 9th and August 31st (included), respectively. *Glitch_k* takes the value one *k* days after the glitch for stocks affected by the glitch and zero otherwise. *DMM_k* takes the value one *k* days after the glitch for stocks whose designated market maker is Knight Capital. The control variables are log of market capitalization, log of price, log of turnover, and volatility computed based on daily high and low prices as in Parkinson (1980). The sample consists of daily observations on NYSE-listed S&P 500 stocks (394 stocks) from January 2012 to December 2012, excluding days with an early closure (07/03, 11/23, and 12/24). All the variables (except indicators) are winsorized at 0.05% and 99.95%. Standard errors are clustered by stock and reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level.

Dependent variable:						
	ES [bp]	ES [bp]	log turn	log turn	depth [\$K]	depth [\$K]
Glitch0	3.26*** (0.73)	3.26*** (0.73)	0.17** (0.08)	0.17** (0.08)	-172.01** (80.22)	-172.03** (80.23)
GlitchNextWeek	-0.13 (0.24)		0.07 (0.04)		-41.54 (45.56)	
Glitch1		-0.53* (0.28)		0.09 (0.06)		-116.89 (91.04)
Glitch2		0.70 (0.60)		0.13** (0.06)		25.13 (66.80)
Glitch3		-0.19 (0.21)		0.10** (0.05)		-21.98 (29.06)
Glitch4		-0.40 (0.25)		0.11* (0.06)		-29.96 (36.06)
Glitch5		-0.25 (0.38)		-0.07 (0.05)		-64.01 (61.00)
GlitchAug9toAug31	0.05 (0.13)	0.05 (0.13)	0.03 (0.04)	0.03 (0.04)	10.64 (24.16)	10.64 (24.16)
DMM0	-0.14 (0.16)	-0.14 (0.16)	0.05 (0.09)	0.05 (0.09)	11.01 (17.22)	11.01 (17.22)
DMMNextWeek	-0.06 (0.14)		-0.02 (0.06)		-6.89 (9.51)	
DMM1		0.15 (0.29)		-0.06 (0.08)		25.73 (19.78)
DMM2		0.38 (0.53)		-0.05 (0.12)		5.87 (13.13)
DMM3		-0.30 (0.21)		0.01 (0.06)		-20.04* (11.57)
DMM4		-0.46 (0.28)		-0.02 (0.09)		-21.74** (9.60)
DMM5		-0.09 (0.28)		0.01 (0.08)		-24.29** (9.95)
DMMAug9toAug31	-0.06 (0.11)	-0.06 (0.11)	-0.04 (0.05)	-0.04 (0.05)	2.22 (7.78)	2.22 (7.78)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	96,552	96,552	96,552	96,552	96,552	96,552
Adjusted R ²	0.17	0.17	0.34	0.34	0.07	0.07

Table 8: Standard liquidity measures around the glitch (extended sample). This table reports the results of regressing daily percentage effective spread, realized spread, price impact, and (log) turnover on indicator variables, day and stock fixed effects, and several control variables. *Glitch_k* takes the value one k days after the glitch for stocks affected by the glitch and zero otherwise. *GlitchAug9ToAug31* is defined in a similar way for glitch-affected stocks between August 9th and August 31st (included). *DMM_k* takes the value one k days after the glitch for stocks whose designated market maker is Knight Capital and who are not affected by the glitch. *DMMAug9ToAug31* is defined in a similar way for stocks whose designated market maker is Knight Capital between August 9th and August 31st (included). The control variables are log of market capitalization, log of price, log of turnover, and volatility computed based on daily high and low prices as in Parkinson (1980). The sample consists of daily observations on NYSE/Amex-listed stocks from July 1, 2012 to September 30, 2012, excluding days with an early closure (07/03). Stocks are required to have a price greater than \$2 and lower than \$1000 at the end of June, a market capitalization greater than \$100 million at the end of June, and valid data for all variables every day of the sample. All the variables (except indicators) are winsorized at 0.05% and 99.95%. Standard errors are clustered by stock and reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% level.

	Effective Spread [bp]	Realized Spread [bp]	Price Impact [bp]	log turn
Glitch0	8.33*** (1.93)	28.01*** (6.49)	-19.10*** (4.81)	0.33*** (0.08)
Glitch1	-1.19** (0.51)	-1.53 (1.01)	0.44 (1.00)	0.25*** (0.05)
Glitch2	-0.36 (0.55)	1.66 (1.15)	-2.00* (1.09)	0.21*** (0.05)
Glitch3	2.29* (1.26)	3.44 (3.07)	-1.09 (1.95)	0.15*** (0.05)
Glitch4	0.26 (0.40)	0.02 (0.61)	0.30 (0.62)	0.10** (0.05)
Glitch5	-1.54** (0.60)	0.69 (1.16)	-2.40** (1.18)	0.05 (0.05)
GlitchAug9toAug31	0.30 (0.25)	-0.13 (0.37)	0.39 (0.47)	0.08*** (0.03)
DMM0	0.37 (1.64)	-2.13 (1.49)	2.41 (2.04)	0.05 (0.05)
DMM1	5.45* (2.87)	1.53 (1.41)	3.88 (2.66)	-0.02 (0.05)
DMM2	1.05 (1.26)	4.17 (2.86)	-2.61 (1.88)	-0.08* (0.05)
DMM3	1.47 (3.21)	-0.09 (1.86)	1.67 (3.29)	0.04 (0.04)
DMM4	-0.11 (1.89)	-0.94 (1.51)	0.75 (1.20)	0.03 (0.04)
DMM5	1.30 (2.47)	-0.32 (1.55)	1.31 (2.47)	-0.09 (0.05)
DMMAug9toAug31	-0.37 (0.63)	0.27 (0.54)	-0.71 (0.77)	-0.01 (0.03)
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes
Observations	107,502	107,502	107,502	107,502
Adjusted R ²	0.06	0.02	0.04	0.29

Table 9: Panel regressions of institutional liquidity measures and matched standard liquidity measures (short window). This table reports the results of regressing cost measures on indicator variables, control variables, and day and stock fixed effects using institutional parent-order executions on NYSE-listed S&P 500 stocks from June 2012 to September 2012. For each institutional parent-order execution, standard liquidity measures and control variables are computed over the interval of the parent-order execution as described in Section 4.4. *GlitchDay0* equals one if the parent-order execution is made on a glitch-affected stock on the day of the glitch. *GlitchNextWeek*, *GlitchAug9ToAug31*, and *GlitchPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. *DMM0* equals one if the parent-order execution is made on a stock for which Knight Capital is the designated market maker on the day of the glitch. *DMMNextWeek*, *DMMAug9ToAug31*, and *DMMPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. The liquidity measures are defined in Section 3. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

<i>Dependent variable:</i>	VWAP Slippage	Implementation Shortfall	Effective Spread	Realized Spread	Price Impact
GlitchDay0	8.13** (3.24)	13.39 (15.09)	1.00** (0.51)	0.38 (0.74)	0.63 (1.04)
GlitchNextWeek	6.48*** (1.96)	19.22** (7.47)	0.35 (0.33)	-0.78 (0.94)	1.14 (0.82)
GlitchAug9ToAug31	1.28* (0.74)	-3.32 (4.62)	0.21 (0.21)	0.27 (0.36)	-0.05 (0.40)
GlitchPost	0.49 (0.95)	-2.96 (4.42)	0.23 (0.19)	0.28 (0.36)	-0.04 (0.41)
DMM0	4.43** (2.11)	24.33 (16.32)	-0.12 (0.19)	-0.30 (0.50)	0.18 (0.47)
DMMNextWeek	-4.05 (3.01)	-1.38 (10.96)	-0.16 (0.30)	-0.004 (1.19)	-0.16 (1.11)
DMMAug9ToAug31	-0.34 (2.52)	-9.40 (11.11)	-0.10 (0.25)	0.54 (0.62)	-0.65 (0.63)
DMMPost	0.27 (1.95)	-8.19 (12.92)	-0.09 (0.26)	0.87 (0.76)	-0.95 (0.82)
Participation Rate	18.92*** (3.26)	31.49*** (11.41)	0.44 (0.83)	-3.56** (1.57)	4.00** (1.73)
Fraction of Daily Volume	25.94** (10.16)	148.61* (83.68)	-5.19*** (1.71)	5.75 (3.64)	-10.94*** (3.99)
Relative Daily Volume	-1.42** (0.57)	1.29 (3.34)	-0.33** (0.16)	-0.18 (0.16)	-0.15 (0.26)
IsEarningsDay	1.28 (1.36)	13.87 (9.15)	-0.19 (0.31)	-0.88 (0.67)	0.70 (0.75)
Interval Turnover	0.49*** (0.15)	0.80 (0.78)	-0.02 (0.02)	0.03 (0.03)	-0.04 (0.04)
Interval Volatility	-69.69 (69.45)	-227.26 (267.61)	249.02** (103.26)	94.59 (59.78)	154.42 (133.93)
Day Volatility	-69.13 (121.21)	-867.85 (580.43)	-20.44 (17.69)	-6.10 (29.59)	-14.33 (38.71)
Execution Duration Fraction	3.28** (1.57)	4.65 (5.87)	0.57** (0.28)	0.24 (0.29)	0.33 (0.36)
Day Return	41.54* (21.25)	24.41 (112.00)	-2.32 (1.93)	-8.68 (5.31)	6.36 (5.76)
abs(Day Return)	18.23 (31.64)	174.96 (154.66)	-7.45 (4.78)	-19.61*** (7.52)	12.17 (9.76)
IsVWAP	-5.04*** (0.73)	-12.38*** (2.19)	-0.06 (0.14)	-0.62** (0.26)	0.57* (0.29)
Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	10,876	10,876	10,828	10,828	10,828
Adjusted R ²	0.14	0.04	0.60	0.04	0.15

Table 10: Panel regressions of institutional liquidity measures and matched standard liquidity measures (extended list of glitch-affected stocks). This table reports the results of regressing cost measures on indicator variables, control variables, and day and stock fixed effects using institutional parent-order executions on NYSE-listed S&P 500 stocks from January 2012 to December 2012. For each institutional parent-order execution, standard liquidity measures and control variables are computed over the interval of the parent-order execution as described in Section 4.4. *GlitchDay0* equals one if the parent-order execution is made on a glitch-affected stock on the day of the glitch. *GlitchNextWeek*, *GlitchAug9ToAug31*, and *GlitchPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. *DMM0* equals one if the parent-order execution is made on a stock for which Knight Capital is the designated market maker on the day of the glitch. *DMMNextWeek*, *DMMAug9ToAug31*, and *DMMPost* are defined in a similar way for parent-order executions made within the next 5 business days after the glitch, between August 9th and August 31st (included), and on any date after August 31st, respectively. The liquidity measures are defined in Section 3. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

<i>Dependent variable:</i>	VWAP Slippage	Implementation Shortfall	Effective Spread	Realized Spread	Price Impact
GlitchDay0	4.80*** (1.52)	19.90*** (6.37)	0.76*** (0.20)	0.21 (0.48)	0.55 (0.58)
GlitchNextWeek	5.11*** (1.68)	16.39*** (6.04)	0.43 (0.27)	-0.54** (0.25)	0.98** (0.39)
GlitchAug9ToAug31	0.16 (0.59)	2.74 (2.20)	0.25 (0.21)	0.06 (0.22)	0.19 (0.31)
GlitchPost	0.11 (0.65)	-0.10 (2.29)	0.23 (0.23)	0.01 (0.22)	0.22 (0.28)
DMM0	4.65*** (0.77)	31.07*** (7.72)	-0.15 (0.10)	-1.04*** (0.34)	0.89** (0.35)
DMMNextWeek	-3.72 (3.38)	3.76 (6.32)	-0.04 (0.22)	-0.52 (0.63)	0.47 (0.50)
DMMAug9ToAug31	-0.02 (1.09)	-3.43 (2.71)	-0.06 (0.19)	0.07 (0.30)	-0.12 (0.41)
DMMPost	-0.003 (0.81)	-3.44 (5.38)	-0.05 (0.16)	0.08 (0.26)	-0.14 (0.31)
Participation Rate	14.78*** (2.17)	35.61*** (6.68)	0.88** (0.42)	-0.84 (1.01)	1.72* (1.02)
Fraction of Daily Volume	25.79*** (6.09)	128.56*** (48.95)	-6.90*** (1.20)	1.80 (2.39)	-8.70*** (2.23)
Relative Daily Volume	-0.27 (0.50)	-0.32 (1.72)	-0.26*** (0.07)	-0.24*** (0.08)	-0.02 (0.10)
IsEarningsDay	0.04 (0.90)	8.74** (4.18)	-0.12 (0.13)	-0.19 (0.31)	0.07 (0.34)
Interval Turnover	0.26*** (0.08)	0.31 (0.40)	-0.01 (0.01)	0.02 (0.01)	-0.03* (0.02)
Interval Volatility	7.72 (30.48)	63.41 (130.39)	200.94*** (45.01)	31.73 (27.71)	169.20*** (55.43)
Day Volatility	-74.74 (60.64)	-828.64*** (301.92)	-17.22 (11.61)	23.32 (21.11)	-40.55 (26.16)
Execution Duration Fraction	2.99*** (0.76)	5.38 (3.59)	0.43*** (0.12)	0.19 (0.17)	0.24 (0.17)
Day Return	20.91* (11.12)	-161.92** (72.17)	-2.13* (1.09)	-6.28** (2.78)	4.15 (2.89)
abs(Day Return)	-7.75 (19.92)	275.24** (118.00)	-2.55 (2.53)	-15.90*** (3.99)	13.35*** (4.79)
IsVWAP	-3.27*** (0.37)	-7.44*** (1.21)	0.004 (0.07)	-0.27* (0.16)	0.27 (0.18)
Fixed Effects	Yes	Yes	Yes	Yes	Yes
Obs.s	31,792	31,792	31,641	31,641	31,641
Adjusted R ²	0.10	0.04	0.61	0.02	0.15

Table 11: Panel regressions of institutional liquidity measures and matched standard liquidity measures (including NASDAQ-listed stocks). This table reports the results of regressing cost measures on indicator variables, control variables, and day and stock fixed effects using institutional parent-order executions on both NYSE-listed and NASDAQ-listed S&P 500 stocks from January 2012 to December 2012. For each institutional parent-order execution, matched standard liquidity measures are computed over the interval of the parent-order execution as described in Section 4.4. The liquidity measures are defined in Section 3. Standard errors are given in parentheses and are double-clustered by day and stock. *, **, and *** denote significance at the 10%, 5%, and 1% level.

<i>Dependent variable:</i>	VWAP Slippage	Implementation Shortfall	Effective Spread	Realized Spread	Price Impact
GlitchDay0	5.65*** (1.72)	13.61** (6.45)	1.04*** (0.25)	-0.52 (0.52)	1.56** (0.64)
GlitchNextWeek	6.32*** (1.56)	14.06** (6.78)	0.40 (0.31)	-1.17 (0.84)	1.57** (0.78)
GlitchAug9ToAug31	1.19** (0.55)	-1.54 (3.25)	0.32 (0.25)	0.05 (0.16)	0.27 (0.31)
GlitchPost	-0.06 (0.55)	-1.39 (2.48)	0.14 (0.27)	-0.02 (0.26)	0.15 (0.32)
DMM0	4.70*** (1.01)	31.07*** (8.36)	-0.03 (0.12)	-1.02*** (0.30)	0.98*** (0.30)
DMMNextWeek	-2.89 (2.89)	2.19 (7.84)	0.01 (0.22)	-0.66 (0.65)	0.67 (0.59)
DMMAug9ToAug31	0.06 (1.04)	-3.13 (2.92)	-0.05 (0.19)	-0.01 (0.32)	-0.04 (0.42)
PostDMM	-0.32 (0.83)	-3.75 (5.39)	-0.12 (0.18)	0.07 (0.24)	-0.19 (0.31)
Participation Rate	12.66*** (1.94)	32.99*** (6.01)	1.04** (0.44)	-1.02 (0.94)	2.07** (0.98)
Fraction of Daily Volume	27.19*** (5.46)	133.66*** (44.63)	-7.46*** (1.22)	-0.17 (2.58)	-7.28*** (2.64)
Relative Daily Volume	-0.47 (0.52)	-0.01 (1.51)	-0.31*** (0.07)	-0.20*** (0.07)	-0.11 (0.10)
IsEarningsDay	-0.55 (0.81)	8.23** (3.70)	-0.12 (0.11)	-0.21 (0.27)	0.09 (0.29)
Interval Turnover	0.31*** (0.09)	-0.09 (0.30)	-0.01** (0.01)	0.01 (0.01)	-0.03** (0.01)
Interval Volatility	9.24 (26.47)	191.74* (113.94)	212.81*** (38.28)	35.23 (24.99)	177.58*** (48.16)
Day Volatility	-59.55 (56.06)	-705.71** (284.71)	-20.15* (10.48)	25.39 (19.49)	-45.55** (23.23)
Execution Duration Fraction	1.94** (0.85)	7.78** (3.31)	0.44*** (0.10)	0.13 (0.16)	0.31** (0.15)
Day Return	8.57 (14.24)	-113.47 (71.45)	-3.14*** (1.16)	-6.94** (2.74)	3.79 (2.74)
abs(Day Return)	16.40 (23.12)	194.03* (110.98)	-3.65 (2.29)	-17.72*** (3.88)	14.08*** (4.48)
IsVWAP	-3.02*** (0.34)	-7.07*** (1.11)	0.06 (0.07)	-0.22 (0.15)	0.28* (0.16)
Fixed Effects	Yes	Yes	Yes	Yes	Yes
Observations	39,173	39,173	39,022	39,022	39,022
Adjusted R ²	0.09	0.04	0.53	0.02	0.13

A. Inventory and Slow-Moving Capital Models

This section lays out simple inventory and slow-moving capital models. More in-depth discussions can be found in [Duffie \(2010\)](#) and [Bogousslavsky \(2016\)](#).

Consider an infinite-horizon one-asset economy with two assets: a risky asset that pays a dividend $d_t \sim N(\bar{d}, \sigma_d)$ every period and a risk-free asset that pays an exogenous gross return R every period.

Three groups of investors trade in the economy. First, risk-averse liquidity providers maximize expected utility over next period wealth, W_{t+1} . The optimization problem of a liquidity provider is given by $\max_{x_{F,t}} -\mathbb{E}_t e^{-\gamma_F W_{t+1}}$, where $W_{t+1} = RW_t + x_{F,t}(p_{t+1} + d_{t+1} - Rp_t)$, where p_t denote the price of the risky asset and γ_F the trader's risk aversion. Second, infrequent traders maximize expected utility over wealth the next time they rebalance their portfolio. For simplicity, the rebalancing frequency is exogenous and equal to $k+1$. Here, k can be interpreted as the inattention period. The optimization problem of an infrequent trader is given by $\max_{x_{I,t}} -\mathbb{E}_t e^{-\gamma_I W_{t+k+1}^I}$, where $W_{t+k+1}^I = R^{k+1}W_t^I + x_{I,t}(p_{t+k+1} + \sum_{j=1}^{k+1} R^{k+1-j}d_{t+j} - R^{k+1}p_t)$. Dividends paid while the trader is out of the market are reinvested at the risk-free rate. Finally, liquidity traders submit a supply shock $\theta_t = a_\theta \theta_{t-1} + \epsilon_{\theta,t}$, where $\epsilon_{\theta,t} \sim N(0, \sigma_\theta)$.

Liquidity providers are present in mass $1 - q$ and infrequent traders are present in mass q . We assume that an equal fraction of infrequent traders rebalance their portfolio every period. The market-clearing condition is then given by $(1 - q)x_{F,t} + \frac{q}{k+1}x_{I,t} = \bar{s} + \theta_t - \sum_{j=1}^k \frac{q}{k+1}x_{I,t-j}$, where $\bar{s}$ is the risky asset's number of shares outstanding.

Standard inventory model ($q = 0$). The standard inventory model is obtained by setting the fraction of infrequent traders to zero. In equilibrium, $p_t = \bar{p} - \lambda \theta_t$, where $R\lambda = \gamma_F(\sigma_d^2 + \lambda^2 \sigma_\theta^2)$. λ is the equilibrium price impact parameter. Whenever the inventory of frequent traders is greater (lower) than zero, the price is below (above) its average value to compensate traders for bearing inventory risk. The price follows a simple autoregressive process, $p_t = \bar{p} + a_\theta(p_{t-1} - \bar{p}) - \lambda \epsilon_{\theta,t}$, and gradually reverts to its average value following a supply shock (e.g., [Figure 2](#)).

Slow-moving capital model ($q > 0$). When a fraction of traders rebalance their portfolio infrequently, equilibrium price dynamics are given by $p_t = \bar{p} - \lambda \theta_t + \sum_{j=1}^k p_{I,j} x_{t-j}^I$. Traders' lagged demands affect equilibrium prices. Since fewer traders are present in the market at any time, supply shocks have a larger price impact than in the standard inventory model. Furthermore, infrequent rebalancing can lead prices to overshoot around the rebalancing horizon (e.g., [Figure 2](#)). Infrequent traders absorb part of the initial supply shock because they know that the price will have recovered when they rebalance. As a result, the lagged demands restrict the available supply of the asset.

Calibration. Figure 2 uses the following calibration: $\bar{d} = 0$; $\bar{s} = 0$; $\gamma_F = \gamma_I = 1$; $a_\theta = 0.5$; $q = 0.6$; $k = 5$; $\sigma_d = 0.1$; $\sigma_\theta = 1$; $R = 1.01$; and a one-unit supply shock. We focus on the positive root (stable equilibrium) of the price impact λ (see Bogouslavsky (2016) for details).

B. History of the Institutional Large Order Execution Data Set

The dataset has been originally obtained in May 2013 for a research project investigating the implications of investor heterogeneity on the estimation of price impact.³⁷ For this purpose, the parent-order executions in the sample have been selected from an active subset of the broker's clients. In this section, we describe in detail how the original data set is constructed.

The raw data set is provided by the Bank in two steps: in-sample and out-of-sample data. The in-sample data were designed with the following filters to estimate the short-term trading skills of a large universe of investors:

- The trading period is from January 2011 to June 2012, inclusive.
- The parent-order is executed via VWAP or POV algorithm.
- The asset universe consists of the S&P 500 stocks.³⁸
- The parent-orders come from active investors only: an investor is considered active if he has at least 100 and at most 500 VWAP or POV orders on S&P 500 stocks between January 2011 to June 2012.
- All orders have been fully filled during regular market hours, opening and closing auctions.
- The parent-order execution duration is greater than 5 minutes but no longer than a full trading day.

These filters are chosen for the purpose of estimating short-term trading skill cleanly. First, VWAP and POV orders are passively managed so the performance of the trade would be mainly based on investor's timing ability rather than the broker's skill. Second, S&P 500 stocks are very liquid and relatively homogeneous subset of the stock universe and it would

³⁷Saglam et al. (2018) is the final version of this research project and use a subset of this dataset.

³⁸Since the data is obtained in May 2013, the traded asset universe in the data set consists of S&P 500 stocks as of May 2013.

be unlikely to have an investor trading with insider information. Third, investors with small number of parent-order executions (< 100) are eliminated as their short-term trading skill may not be estimated reliably with fewer observations. Investors with high number of parent-order executions (> 500) are also filtered to have a balanced data set across clients and prevent any specific investor from fully driving the estimation procedure. Finally, parent-order executions lasting less than 5 minutes are often very small in size that may get executed with a single market order.

With these criteria, the in-sample data consists of 63,379 parent-order executions coming from 293 clients. The Bank then constructed the out-of-sample data set using the similar set of filters but restricting the data to the same set of clients:

- The client submitting the order is one of 293 clients present in the in-sample data.
- The trading period is from July 2012 to December 2012, inclusive.
- The asset universe consists of the S&P 500 stocks.
- All orders have been fully filled during regular market hours, opening and closing auctions.
- The parent-order execution duration is greater than 5 minutes but no longer than a full trading day.

The out-of-sample data set consists of 18,365 parent-order executions coming from 223 clients. Merging both data sets, we have access to 81,744 parent-order executions submitted by 293 unique investors between January 2011 and December 2012.

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